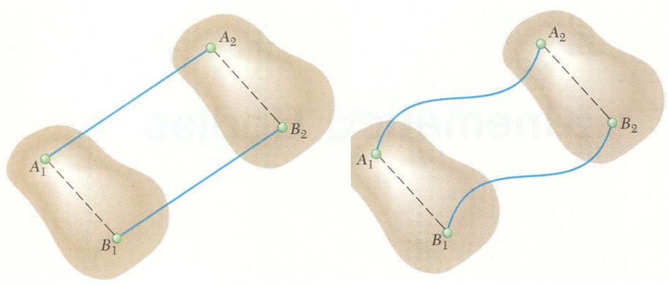
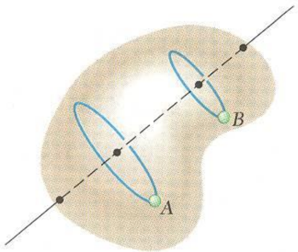
KINEMATICS OF RIGID BODIES

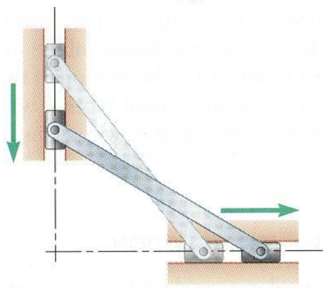
LECTURE: 31-33

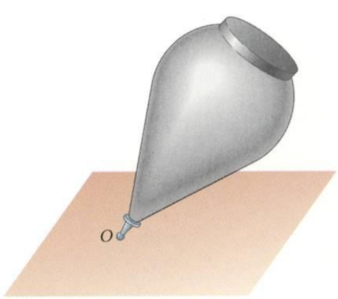
Contents

Introduction



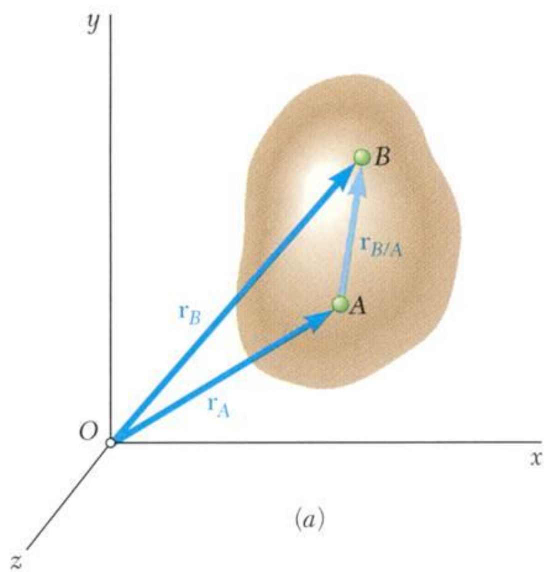


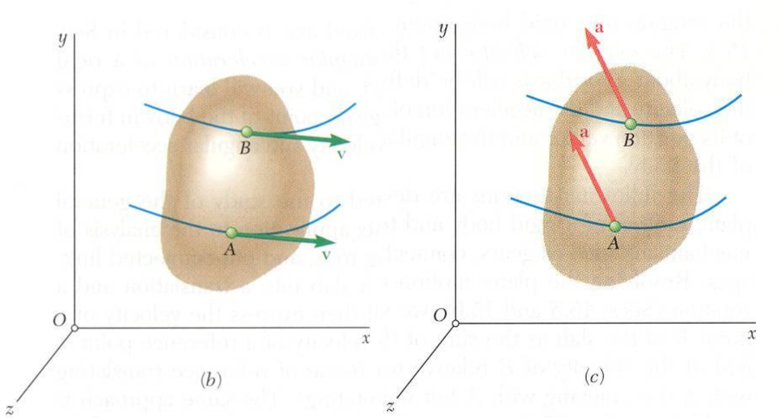




- Kinematics of rigid bodies: relations between time and the positions, velocities, and accelerations of the particles forming a rigid body.
- Classification of rigid body motions:
  - translation:
    - rectilinear translation
    - curvilinear translation
  - rotation about a fixed axis
  - general plane motion
  - motion about a fixed point
  - general motion

Translation





- Consider rigid body in translation:
  - direction of any straight line inside the body is constant,
  - all particles forming the body move in parallel lines.
- For any two particles in the body,

$$\vec{r}_B = \vec{r}_A + \vec{r}_{B/A} \quad (\vec{r}_{B/A} \text{ being constant})$$

- Differentiating with respect to time,

$$\dot{\vec{r}}_B = \dot{\vec{r}}_A + \dot{\vec{r}}_{B/A} = \dot{\vec{r}}_A$$

$$\vec{v}_B = \vec{v}_A$$

All particles have the same velocity.

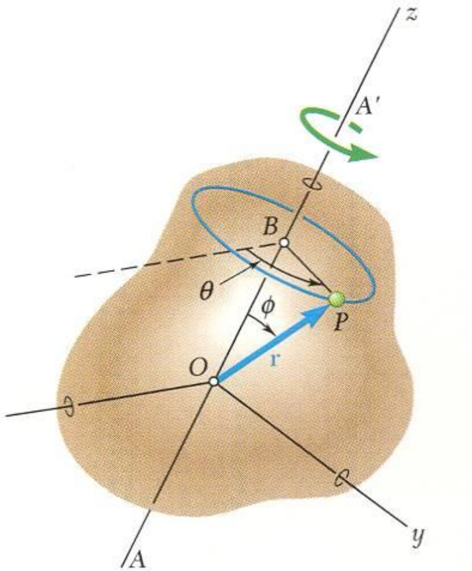
- Differentiating with respect to time again,

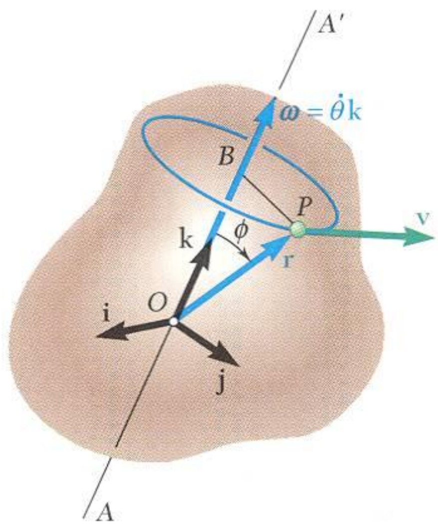
$$\ddot{\vec{r}}_B = \ddot{\vec{r}}_A + \ddot{\vec{r}}_{B/A} = \ddot{\vec{r}}_A$$

$$\vec{a}_B = \vec{a}_A$$

All particles have the same acceleration.

Rotation About a Fixed Axis. Velocity





- Consider rotation of rigid body about a fixed axis *AA*′

- Velocity vector $\vec{v} = d\vec{r}/dt$ of the point *P* is tangent to the blue circle with magnitude $v = ds/dt$

$$\Delta s = (BP)\Delta\theta = (r\sin\phi)\Delta\theta$$

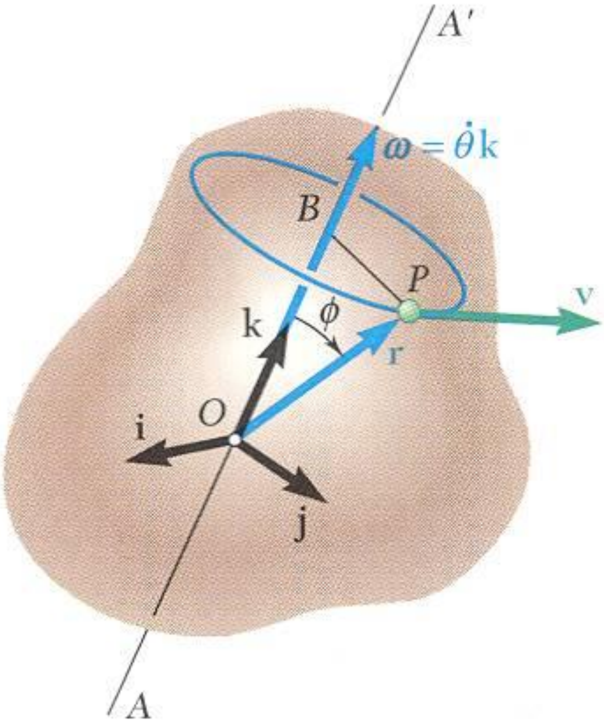
$$v = \frac{ds}{dt} = \lim_{\Delta t \to 0}(r\sin\phi)\frac{\Delta\theta}{\Delta t} = r\dot{\theta}\sin\phi$$

- The same result is obtained from

$$\vec{v} = \frac{d\vec{r}}{dt} = \vec{\omega}\times\vec{r}$$

$$\vec{\omega} = \omega\vec{k} = \dot{\theta}\vec{k} = angular\ velocity$$

Rotation About a Fixed Axis.: Acceleration



- Differentiating to determine the acceleration,

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(\vec{\omega} \times \vec{r})$$
$$= \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$$
$$= \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \vec{v}$$

- $\frac{d\vec{\omega}}{dt} = \vec{\alpha} = angular\ acceleration$

$$= \alpha\vec{k} = \dot{\omega}\vec{k} = \ddot{\theta}\vec{k}$$

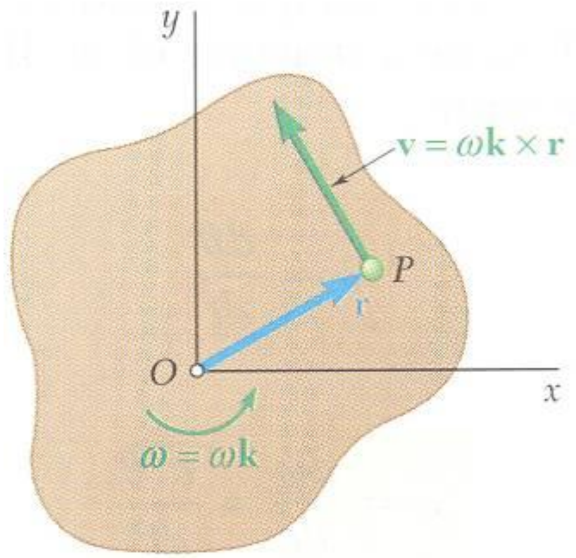
- Acceleration of P is combination of two vectors,

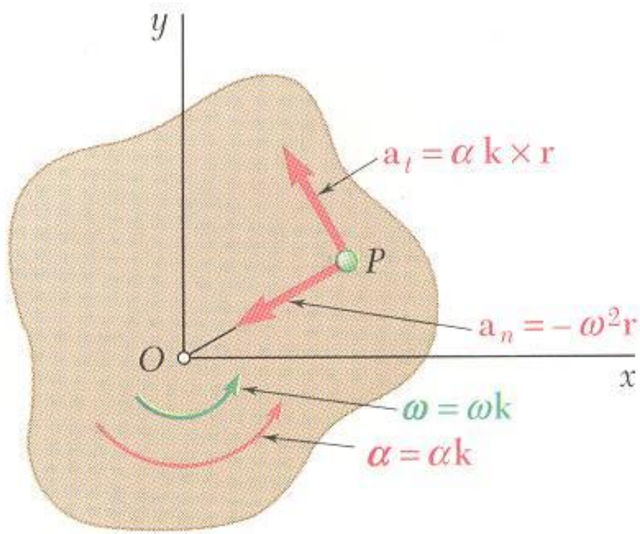
$$\vec{a} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$\vec{\alpha} \times \vec{r} =$ tangential acceleration component

$\vec{\omega} \times (\vec{\omega} \times \vec{r}) =$ radial acceleration component

Rotation About a Fixed Axis.: Representative Slab





- Consider the motion of a representative slab in a plane perpendicular to the axis of rotation.

- Velocity of any point *P* of the slab,

$$\vec{v} = \vec{\omega} \times \vec{r} = \omega\vec{k} \times \vec{r}$$

$$v = r\omega$$

- Acceleration of any point *P* of the slab,

$$\vec{a} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{\omega} \times \vec{r}$$

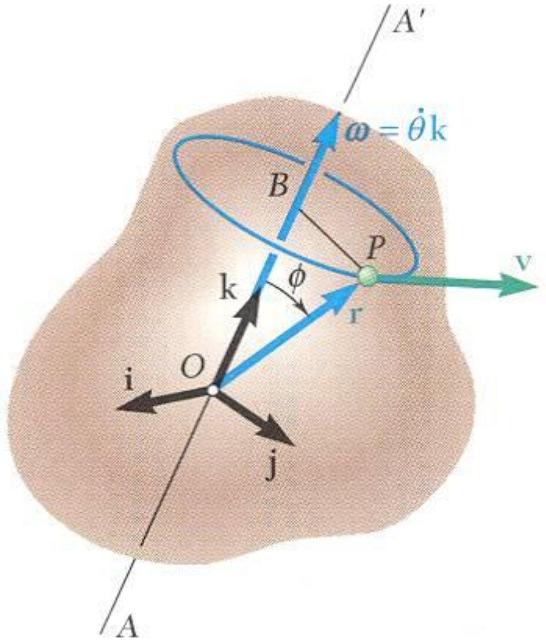
$$= \alpha\vec{k} \times \vec{r} - \omega^2\vec{r}$$

- Resolving the acceleration into tangential and normal components,

$$\vec{a}_t = \alpha\vec{k} \times \vec{r} \qquad a_t = r\alpha$$

$$\vec{a}_n = -\omega^2\vec{r} \qquad a_n = r\omega^2$$

Equations Defining the Rotation of a Rigid Body About a Fixed Axis



- Motion of a rigid body rotating around a fixed axis is often specified by the type of angular acceleration.

- Recall $\omega = \dfrac{d\theta}{dt}$ or $dt = \dfrac{d\theta}{\omega}$

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} = \omega\frac{d\omega}{d\theta}$$

- *Uniform Rotation*, $\alpha = 0$:

$$\theta = \theta_0 + \omega t$$

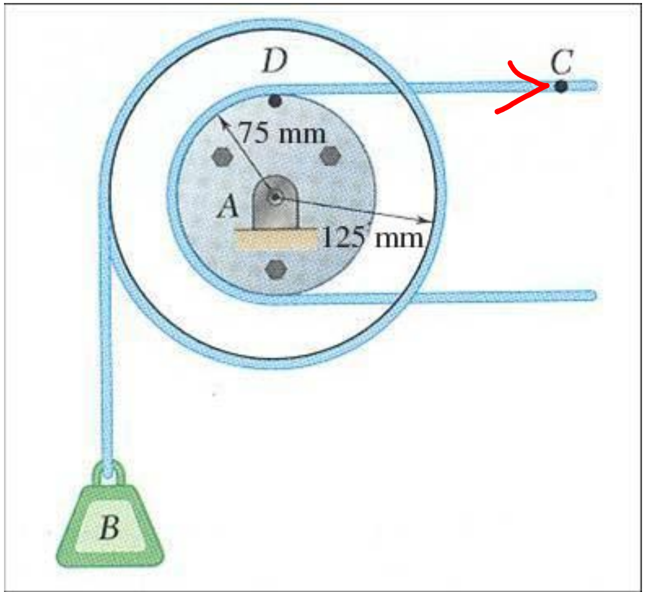
- *Uniformly Accelerated Rotation*, α = constant:

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \tfrac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$$

Sample Problem 1



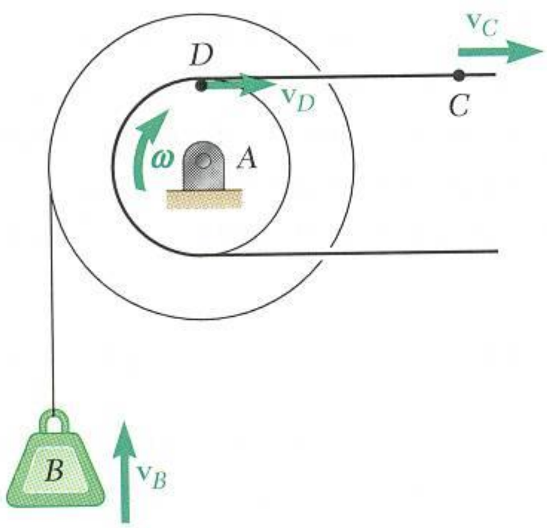
Cable *C* has a constant acceleration of 225 m/s^2 and an initial velocity of 300 mm/s, both directed to the right.

Determine *(a)* the number of revolutions of the pulley in 2 s, *(b)* the velocity and change in position of the load *B* after 2 s, and *(c)* the acceleration of the point *D* on the rim of the inner pulley at $t = 0$.

SOLUTION:

- Due to the action of the cable, the tangential velocity and acceleration of *D* are equal to the velocity and acceleration of *C*. Calculate the initial angular velocity and acceleration.
- Apply the relations for uniformly accelerated rotation to determine the velocity and angular position of the pulley after 2 s.
- Evaluate the initial tangential and normal acceleration components of *D*.

Sample Problem 1



<u>SOLUTION</u>:

- The tangential velocity and acceleration of D are equal to the velocity and acceleration of C.

$$(\vec{v}_D)_0 = (\vec{v}_C)_0 = 300\,\mathrm{mm/s} \rightarrow \qquad (\vec{a}_D)_t = \vec{a}_C = 225\,\mathrm{mm/s^2} \rightarrow$$

$$(v_D)_0 = r\omega_0 \qquad (a_D)_t = r\alpha$$

$$\omega_0 = \frac{(v_D)_0}{r} = \frac{300}{75} = 4\,\mathrm{rad/s}\,\boldsymbol{i} \qquad \alpha = \frac{(a_D)_t}{r} = \frac{225}{75} = 3\,\mathrm{rad/s^2}\,\boldsymbol{i}$$

- Apply the relations for uniformly accelerated rotation to determine velocity and angular position of pulley after 2 s.

$$\omega = \omega_0 + \alpha t = 4\,\mathrm{rad/s} + \left(3\,\mathrm{rad/s^2}\right)(2\,\mathrm{s}) = 10\,\mathrm{rad/s}$$

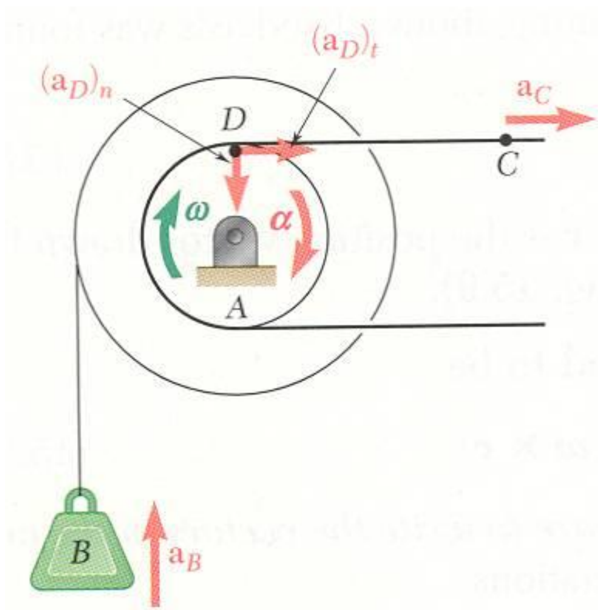
$$\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2 = (4\,\mathrm{rad/s})(2\ \mathrm{s}) + \tfrac{1}{2}\left(3\,\mathrm{rad/s^2}\right)(2\ \mathrm{s})^2$$
$$= 14\ \mathrm{radii}$$

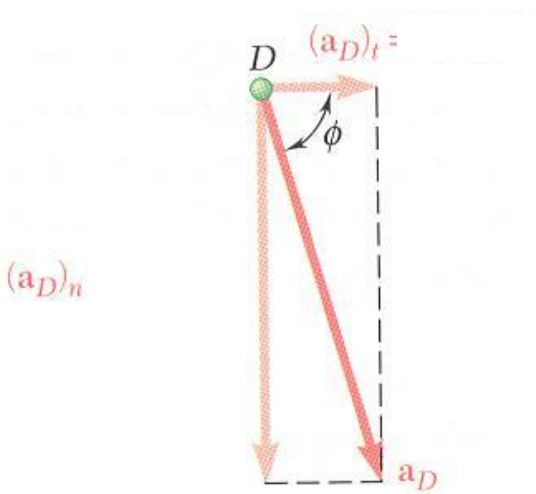
$$N = (14\,\mathrm{rad})\left(\frac{1\,\mathrm{rev}}{2\pi\,\mathrm{rad}}\right) = \text{number of revs} \qquad \boxed{N = 2.23\,\mathrm{rev}}$$

$$v_B = r\omega = (125\ \mathrm{mm})(10\,\mathrm{rad/s}) \qquad \vec{v}_B = 1.25\,\mathrm{m/s}\uparrow$$

$$\Delta y_B = r\theta = (125\ \mathrm{mm})(14\ \mathrm{rad}) \qquad \Delta y_B = 1.75\ \mathrm{m}\uparrow$$

Sample Problem 1





- Evaluate the initial tangential and normal acceleration components of D.

$$\left(\vec{a}_D\right)_t = \vec{a}_C = 225\,\text{mm/s}^2 \rightarrow$$

$$\left(a_D\right)_n = r_D\,\omega_0^2 = (75\,\text{mm})(4\,\text{rad/s})^2 = 1200\,\text{mm/s}^2$$

$$\boxed{\left(\vec{a}_D\right)_t = 225\,\text{mm/s}^2 \rightarrow \qquad \left(\vec{a}_D\right)_n = 1200\,\text{mm/s}^2 \downarrow}$$

Magnitude and direction of the total acceleration,

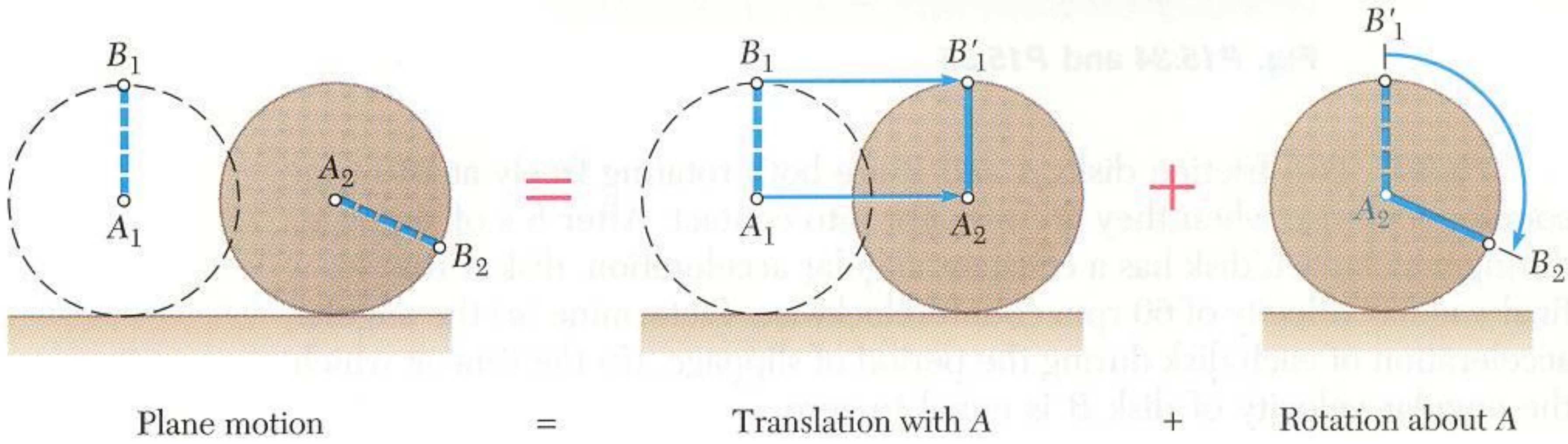
$$\begin{aligned} a_D &= \sqrt{\left(a_D\right)_t^2 + \left(a_D\right)_n^2} \\ &= \sqrt{(225)^2 + (1200)^2} \\ &= 1220.9\,\text{mm/s}^2 \end{aligned}$$

$$\boxed{a_D = 1.221\,\text{m/s}^2}$$

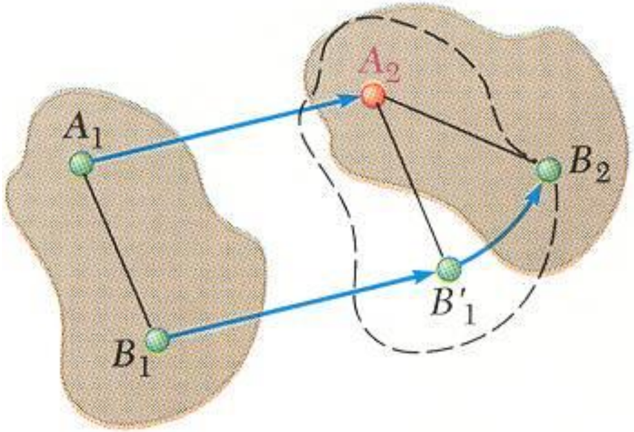
$$\begin{aligned} \tan\phi &= \frac{\left(a_D\right)_n}{\left(a_D\right)_t} \\ &= \frac{1200}{225} \end{aligned}$$

$$\boxed{\phi = 79.4^\circ}$$

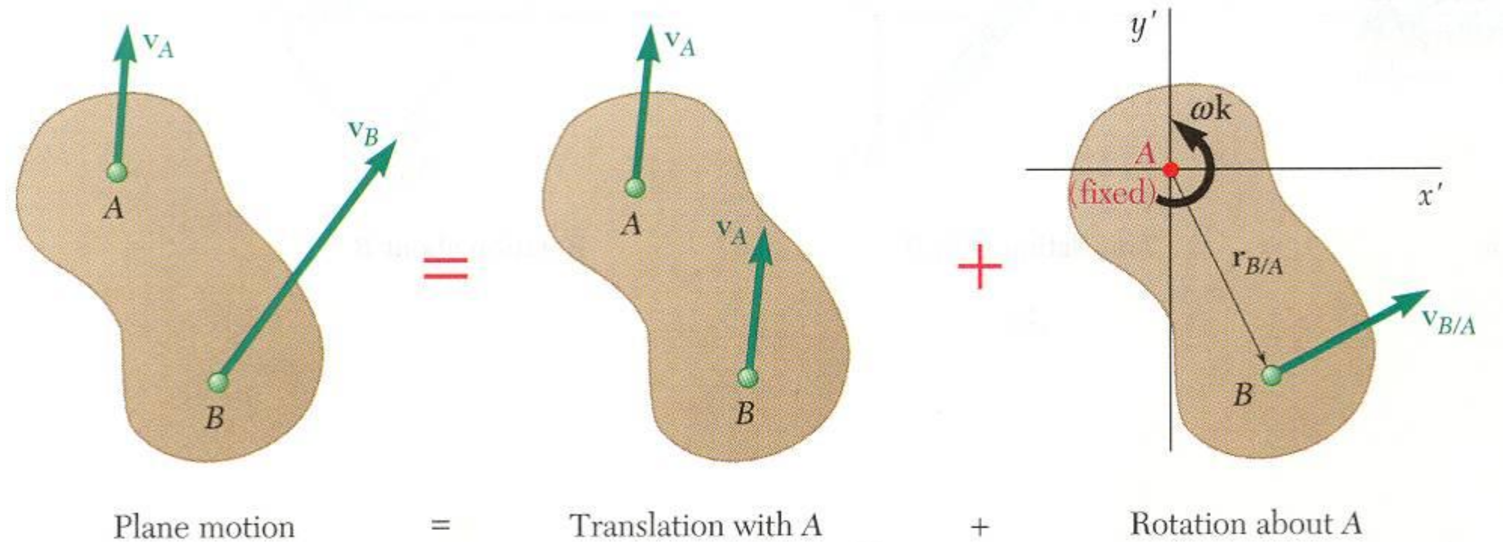
General Plane Motion

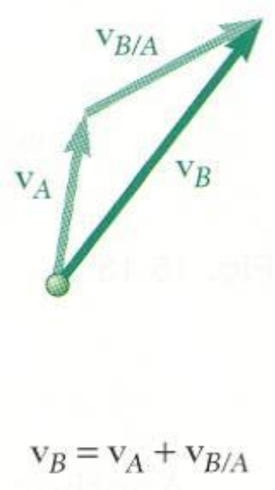


- *General plane motion* is neither a translation nor a rotation.
- General plane motion can be considered as the *sum* of a translation and rotation.
- Displacement of particles A and B to A_2 and B_2 can be divided into two parts:
  - translation to A_2 and B_1'
  - rotation of B_1' about A_2 to B_2



Absolute and Relative Velocity in Plane Motion





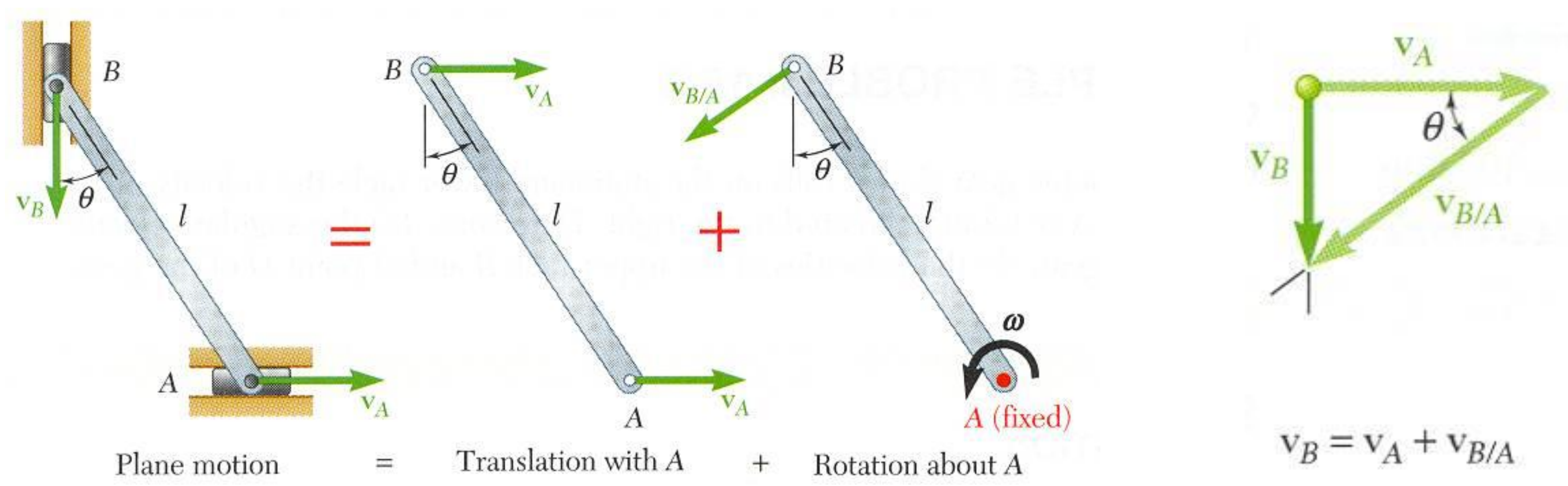
- Any plane motion can be replaced by a translation of an arbitrary reference point *A* and a simultaneous rotation about *A*.

$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$$

$$\vec{v}_{B/A} = \omega\vec{k} \times \vec{r}_{B/A} \qquad v_{B/A} = r\omega$$

$$\vec{v}_B = \vec{v}_A + \omega\vec{k} \times \vec{r}_{B/A}$$

Absolute and Relative Velocity in Plane Motion



- Assuming that the velocity v_A of end A is known, wish to determine the velocity v_B of end B and the angular velocity ω in terms of v_A, l, and θ.
- The direction of v_B and $v_{B/A}$ are known. Complete the velocity diagram.

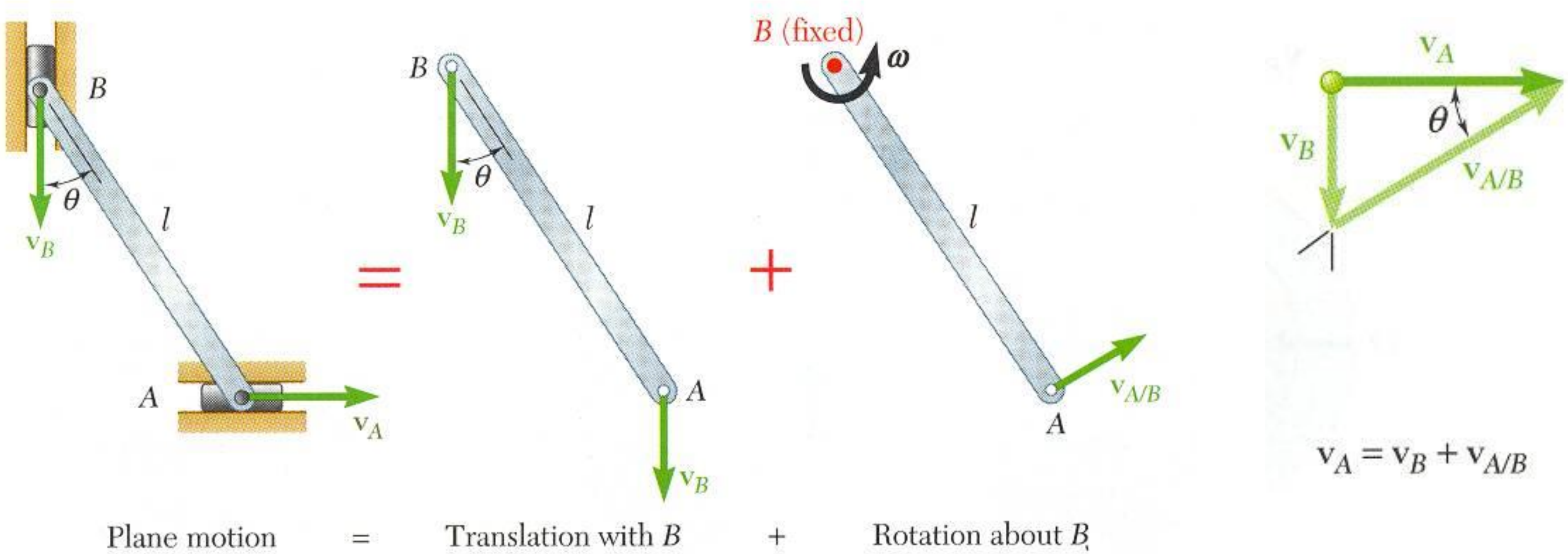
$$\frac{v_B}{v_A} = \tan\theta$$

$$v_B = v_A \tan\theta$$

$$\frac{v_A}{v_{B/A}} = \frac{v_A}{l\omega} = \cos\theta$$

$$\omega = \frac{v_A}{l\cos\theta}$$

Absolute and Relative Velocity in Plane Motion

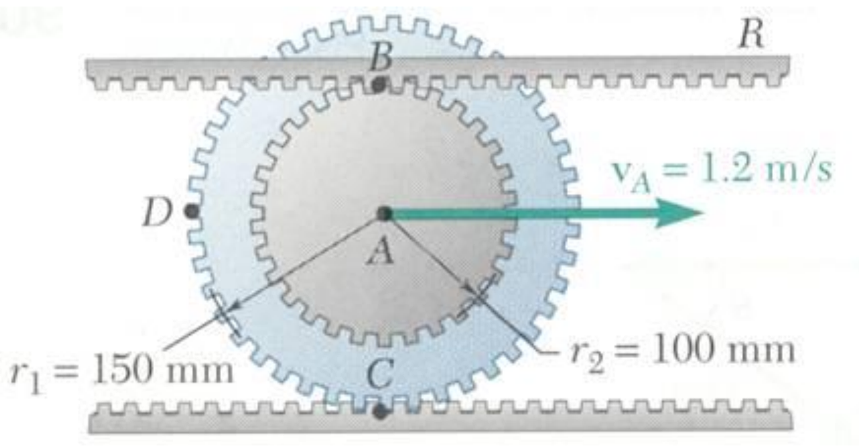


- Selecting point *B* as the reference point and solving for the velocity v_A of end *A* and the angular velocity ω leads to an equivalent velocity triangle.

- $v_{A/B}$ has the same magnitude but opposite sense of $v_{B/A}$. The sense of the relative velocity is dependent on the choice of reference point.

- Angular velocity ω of the rod in its rotation about *B* is the same as its rotation about *A*. Angular velocity is not dependent on the choice of reference point.

Sample Problem 2



The double gear rolls on the stationary lower rack: the velocity of its center is 1.2 m/s.

Determine *(a)* the angular velocity of the gear, and *(b)* the velocities of the upper rack *R* and point *D* of the gear.

SOLUTION:

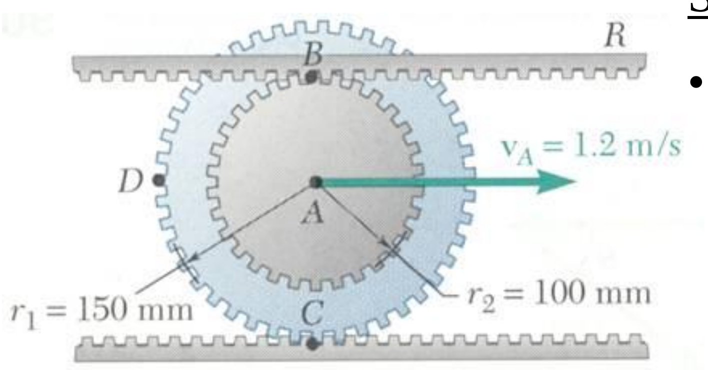
- The displacement of the gear center in one revolution is equal to the outer circumference. Relate the translational and angular displacements. Differentiate to relate the translational and angular velocities.

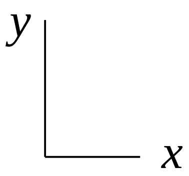
- The velocity for any point *P* on the gear may be written as

$$\vec{v}_P = \vec{v}_A + \vec{v}_{P/A} = \vec{v}_A + \omega\vec{k} \times \vec{r}_{P/A}$$

Evaluate the velocities of points *B* and *D*.

Sample Problem 2





SOLUTION:

- The displacement of the gear center in one revolution is equal to the outer circumference.

For $x_A > 0$ (moves to right), $\omega < 0$ (rotates clockwise).

$$\frac{x_A}{2\pi r_1} = -\frac{\theta}{2\pi} \qquad x_A = -r_1\theta$$

Differentiate to relate the translational and angular velocities.

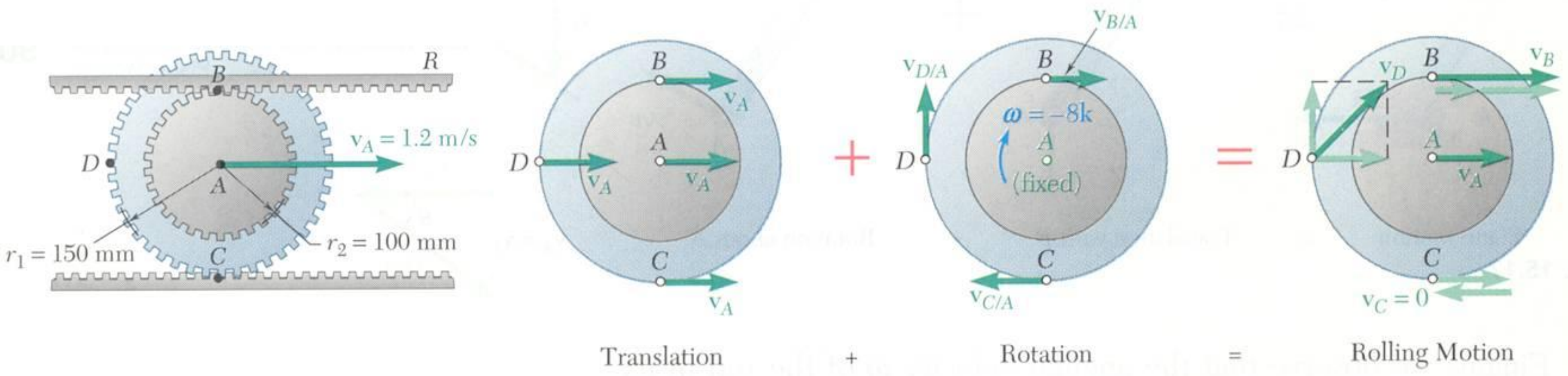
$$v_A = -r_1\omega$$

$$\omega = -\frac{v_A}{r_1} = -\frac{1.2\,\mathrm{m/s}}{0.150\,\mathrm{m}}$$

$$\boxed{\vec{\omega} = \omega\vec{k} = -(8\,\mathrm{rad/s})\vec{k}}$$

Sample Problem 2

- For any point P on the gear, $\vec{v}_P = \vec{v}_A + \vec{v}_{P/A} = \vec{v}_A + \omega\vec{k} \times \vec{r}_{P/A}$



Velocity of the upper rack is equal to velocity of point B:

$$\vec{v}_R = \vec{v}_B = \vec{v}_A + \omega\vec{k} \times \vec{r}_{B/A}$$

$$= 1.2\vec{\imath} + \left(-8\vec{k} \times 0.1\vec{\jmath}\right) \qquad \text{m/s}$$

$$= (1.2\,\text{m/s})\vec{i} + (0.8\,\text{m/s})\vec{i}$$

$$\boxed{\vec{v}_R = (2\,\text{m/s})\vec{i}}$$

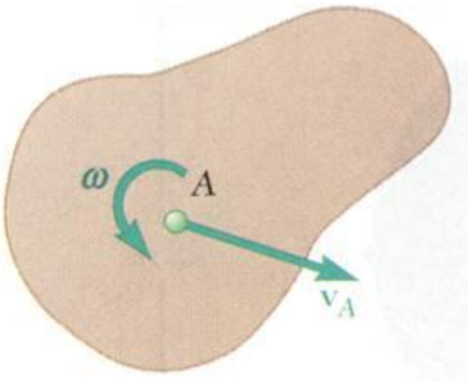
Velocity of the point D:

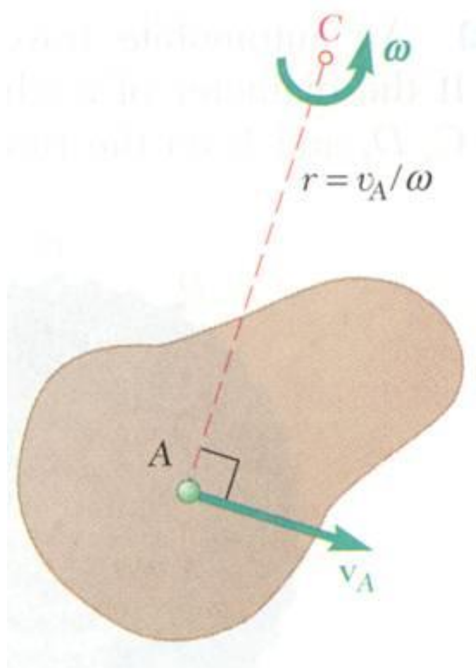
$$\vec{v}_D = \vec{v}_A + \omega\vec{k} \times \vec{r}_{D/A}$$

$$= 1.2\vec{\imath} + \left(-8\vec{k} \times -0.15\vec{\imath}\right) \qquad \text{m/s}$$

$$\boxed{\begin{aligned} \vec{v}_D &= (1.2\,\text{m/s})\vec{i} + (1.2\,\text{m/s})\vec{j} \\ v_D &= 1.697\,\text{m/s} \end{aligned}}$$

Instantaneous Center of Rotation in Plane Motion





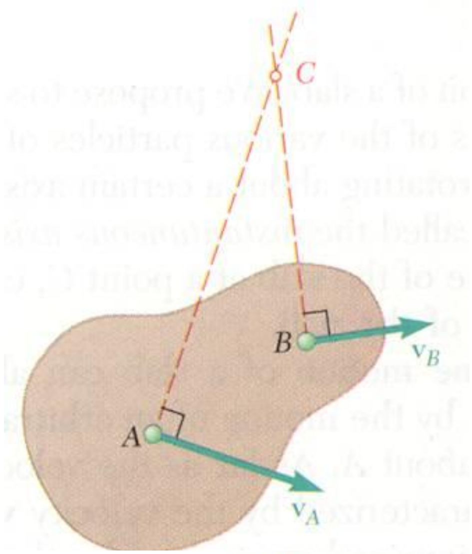
- Plane motion of all particles in a slab can always be replaced by the translation of an arbitrary point *A* and a rotation about *A* with an angular velocity that is independent of the choice of *A*.

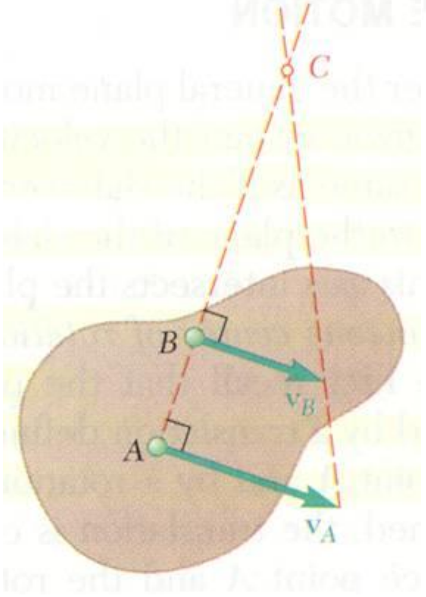
- The same translational and rotational velocities at *A* are obtained by allowing the slab to rotate with the same angular velocity about the point *C* on a perpendicular to the velocity at *A*.

- The velocity of all other particles in the slab are the same as originally defined since the angular velocity and translational velocity at *A* are equivalent.

- As far as the velocities are concerned, the slab seems to rotate about the *instantaneous center of rotation C*.

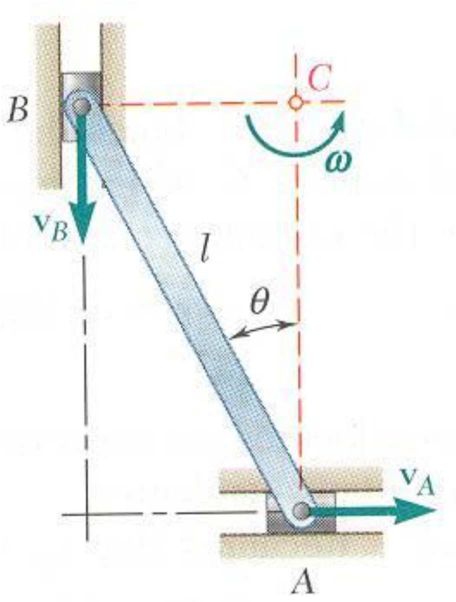
Instantaneous Center of Rotation in Plane Motion





- If the velocity at two points *A* and *B* are known, the instantaneous center of rotation lies at the intersection of the perpendiculars to the velocity vectors through *A* and *B* .
- If the velocity vectors at *A* and *B* are perpendicular to the line *AB*, the instantaneous center of rotation lies at the intersection of the line *AB* with the line joining the extremities of the velocity vectors at *A* and *B.*
- If the velocity vectors are parallel at A and B and they are not perpendicular to line AB, the instantaneous center of rotation is at infinity and the angular velocity is zero.
- If the velocity vectors at *A* and *B* are perpendicular to the line *AB* but the velocity magnitudes are equal, the instantaneous center of rotation is at infinity and the angular velocity is zero.

Instantaneous Center of Rotation in Plane Motion

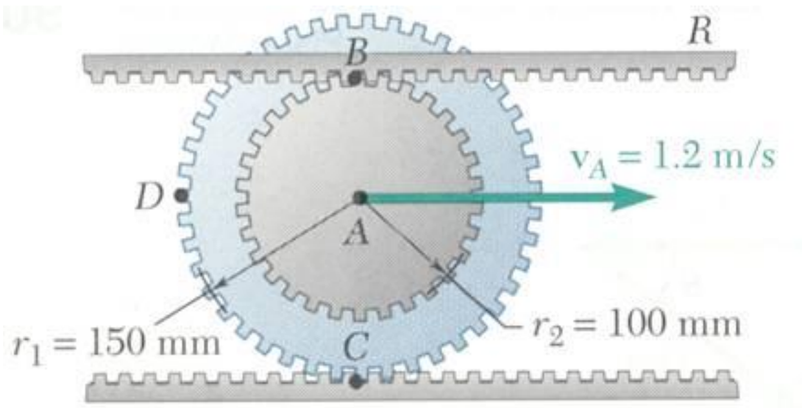


- The instantaneous center of rotation lies at the intersection of the perpendiculars to the velocity vectors through A and B .

$$\omega = \frac{v_A}{AC} = \frac{v_A}{l\cos\theta} \qquad v_B = (BC)\omega = (l\sin\theta)\frac{v_A}{l\cos\theta}$$
$$= v_A \tan\theta$$

- The velocities of all particles on the rod are as if they were rotated about C.
- The particle at the center of rotation has zero velocity.

Sample Problem 3



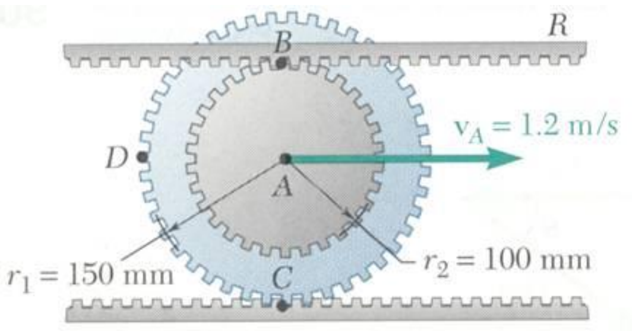
The double gear rolls on the stationary lower rack: the velocity of its center is 1.2 m/s.

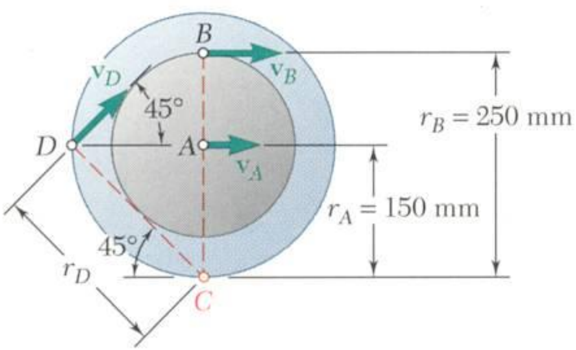
Determine *(a)* the angular velocity of the gear, and *(b)* the velocities of the upper rack *R* and point *D* of the gear.

<u>SOLUTION</u>:

- The point *C* is in contact with the stationary lower rack and, instantaneously, has zero velocity. It must be the location of the instantaneous center of rotation.
- Determine the angular velocity about *C* based on the given velocity at *A*.
- Evaluate the velocities at *B* and *D* based on their rotation about *C*.

Sample Problem 3





SOLUTION:

- The point C is in contact with the stationary lower rack and, instantaneously, has zero velocity. It must be the location of the instantaneous center of rotation.
- Determine the angular velocity about C based on the given velocity at A.

$$v_A = r_A \omega \qquad \omega = \frac{v_A}{r_A} = \frac{1.2\,\mathrm{m/s}}{0.15\,\mathrm{m}} = 8\,\mathrm{rad/s}$$

- Evaluate the velocities at B and D based on their rotation about C.

$$v_R = v_B = r_B \omega = (0.25\,\mathrm{m})(8\,\mathrm{rad/s})$$

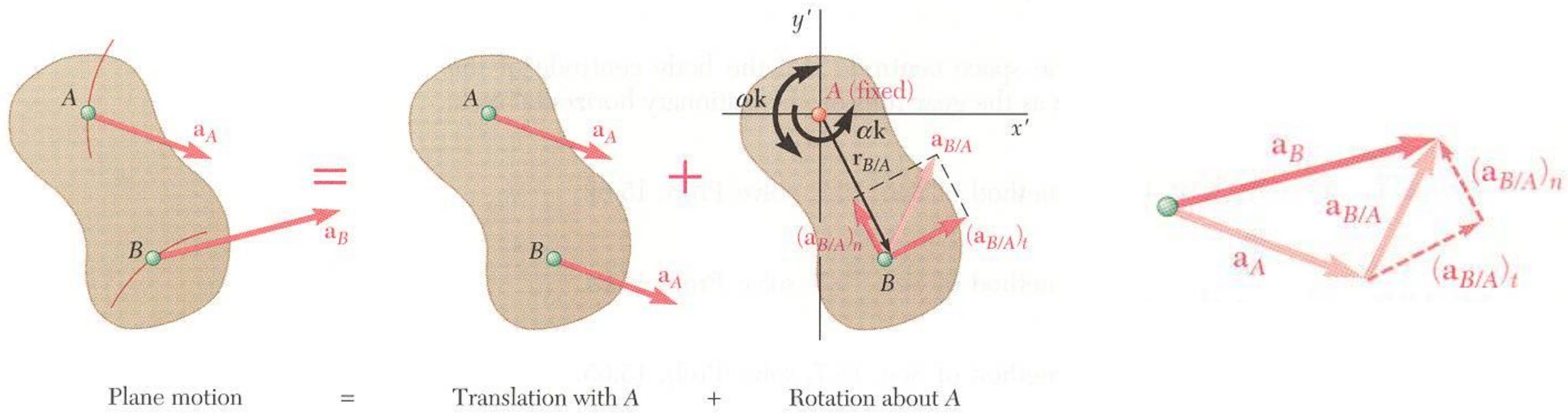
$$\boxed{\vec{v}_R = (2\,\mathrm{m/s})\vec{i}}$$

$$r_D = (0.15\,\mathrm{m})\sqrt{2} = 0.2121\,\mathrm{m}$$

$$v_D = r_D \omega = (0.2121\,\mathrm{m})(8\,\mathrm{rad/s})$$

$$\boxed{\begin{aligned} v_D &= 1.697\,\mathrm{m/s} \\ \vec{v}_D &= (1.2\vec{i} + 1.2\vec{j})(\mathrm{m/s}) \end{aligned}}$$

Absolute and Relative Acceleration in Plane Motion



- Absolute acceleration of a particle of the slab,

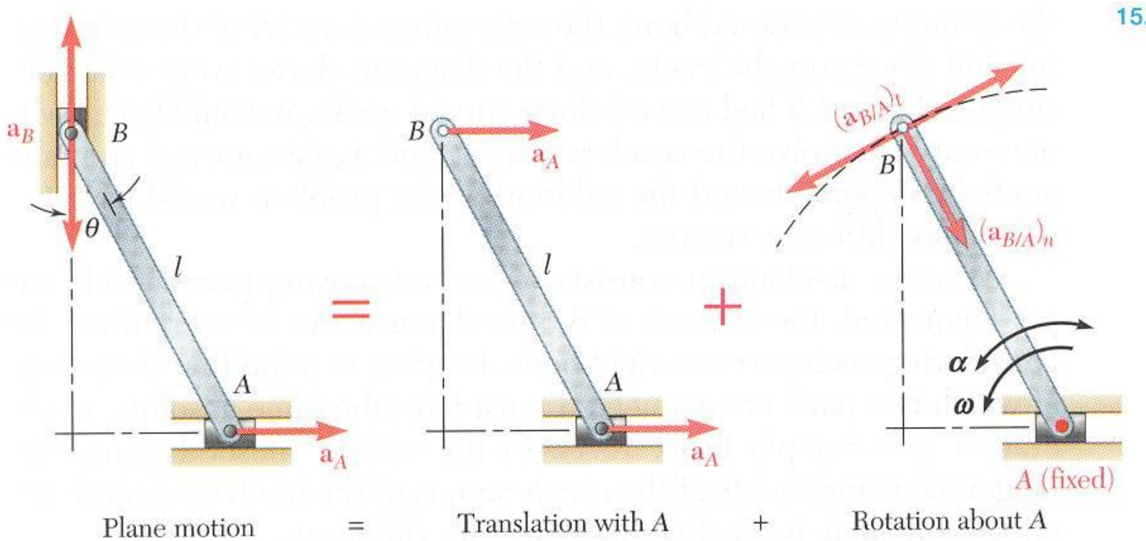
$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$$

- Relative acceleration $\vec{a}_{B/A}$ associated with rotation about A includes tangential and normal components,

$$\left(\vec{a}_{B/A}\right)_t = \alpha\vec{k} \times \vec{r}_{B/A} \qquad \left(a_{B/A}\right)_t = r\alpha$$

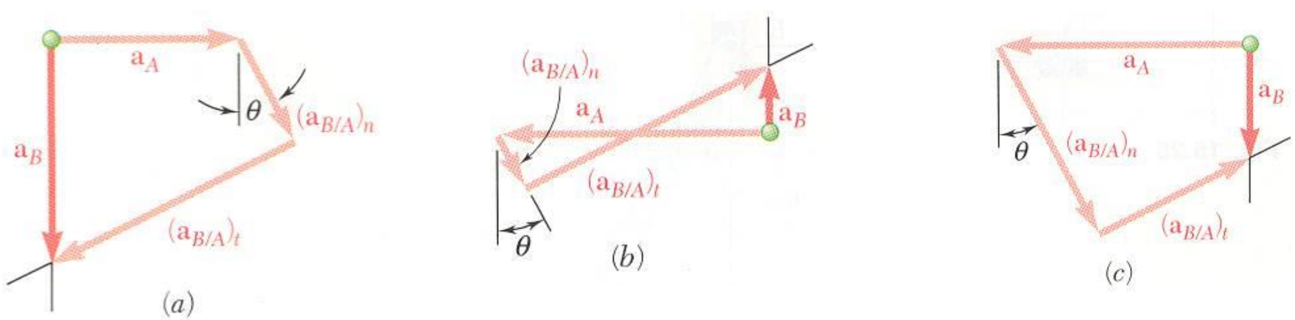
$$\left(\vec{a}_{B/A}\right)_n = -\omega^2 \vec{r}_{B/A} \qquad \left(a_{B/A}\right)_n = r\omega^2$$

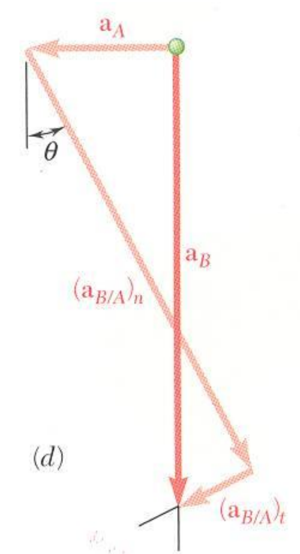
Absolute and Relative Acceleration in Plane Motion



- Given $\vec{a}_A$ and $\vec{v}_A$, determine $\vec{a}_B$ and $\vec{\alpha}$.

$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$$
$$= \vec{a}_A + \left(\vec{a}_{B/A}\right)_n + \left(\vec{a}_{B/A}\right)_t$$

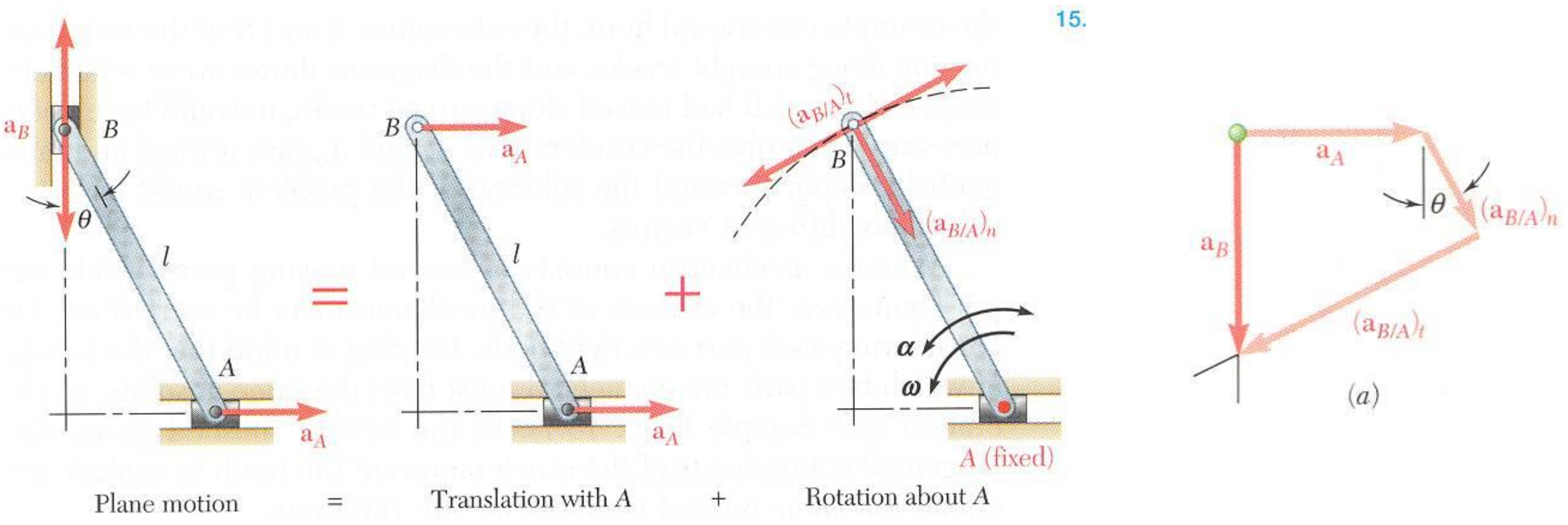




- Vector result depends on sense of $\vec{a}_A$ and the relative magnitudes of a_A and $\left(a_{B/A}\right)_n$

- Must also know angular velocity ω.

Absolute and Relative Acceleration in Plane Motion



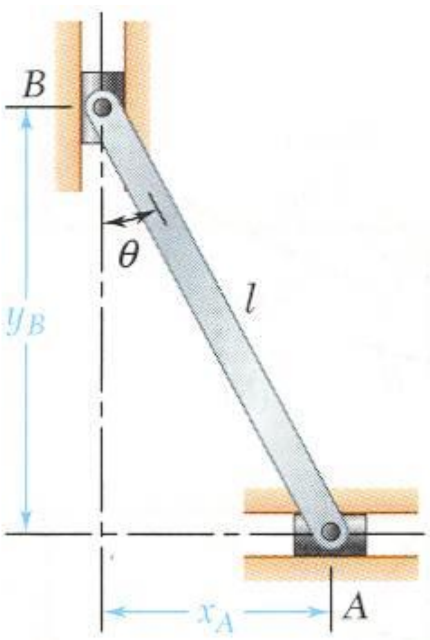
- Write $\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$ in terms of the two component equations,

$$\overset{+}{\rightarrow} \quad x \text{ components:} \qquad 0 = a_A + l\omega^2 \sin\theta - l\alpha\cos\theta$$

$$+\uparrow \quad y \text{ components:} \quad -a_B = -l\omega^2 \cos\theta - l\alpha\sin\theta$$

- Solve for a_B and α.

Analysis of Plane Motion in Terms of a Parameter



- In some cases, it is advantageous to determine the absolute velocity and acceleration of a mechanism directly.

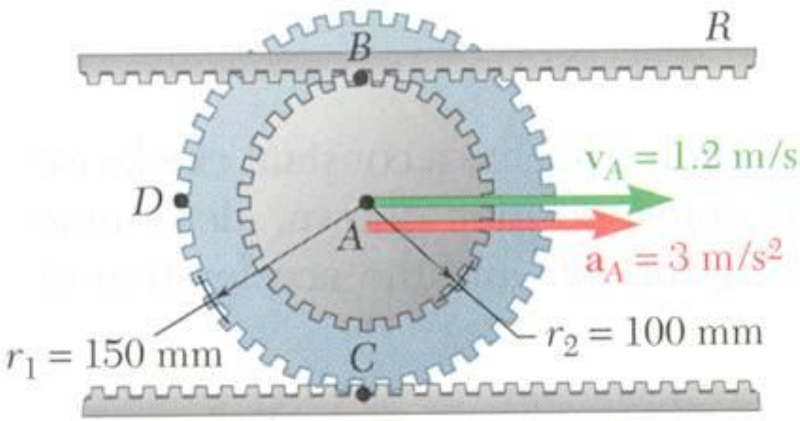
$$x_A = l\sin\theta \qquad\qquad y_B = l\cos\theta$$

$$v_A = \dot{x}_A = l\dot{\theta}\cos\theta = l\omega\cos\theta \qquad\qquad v_B = \dot{y}_B = -l\dot{\theta}\sin\theta = -l\omega\sin\theta$$

$$a_A = \ddot{x}_A = -l\dot{\theta}^2\sin\theta + l\ddot{\theta}\cos\theta = -l\omega^2\sin\theta + l\alpha\cos\theta$$

$$a_B = \ddot{y}_B = -l\dot{\theta}^2\cos\theta - l\ddot{\theta}\sin\theta = -l\omega^2\cos\theta - l\alpha\sin\theta$$

Sample Problem 4



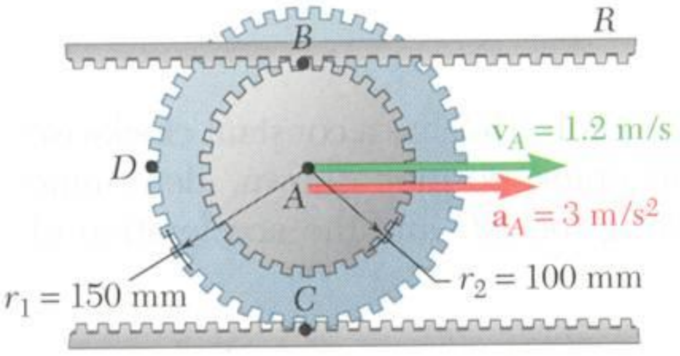
The center of the double gear has a velocity and acceleration to the right of 1.2 m/s and 3 m/s^2, respectively. The lower rack is stationary.

Determine *(a)* the angular acceleration of the gear, and *(b)* the acceleration of points *B*, *C*, and *D*.

SOLUTION:

- The expression of the gear position as a function of θ is differentiated twice to define the relationship between the translational and angular accelerations.
- The acceleration of each point on the gear is obtained by adding the acceleration of the gear center and the relative accelerations with respect to the center. The latter includes normal and tangential acceleration components.

Sample Problem 4



<u>SOLUTION</u>:

- The expression of the gear position as a function of θ is differentiated twice to define the relationship between the translational and angular accelerations.

$$x_A = -r_1\theta$$

$$v_A = -r_1\dot{\theta} = -r_1\omega$$

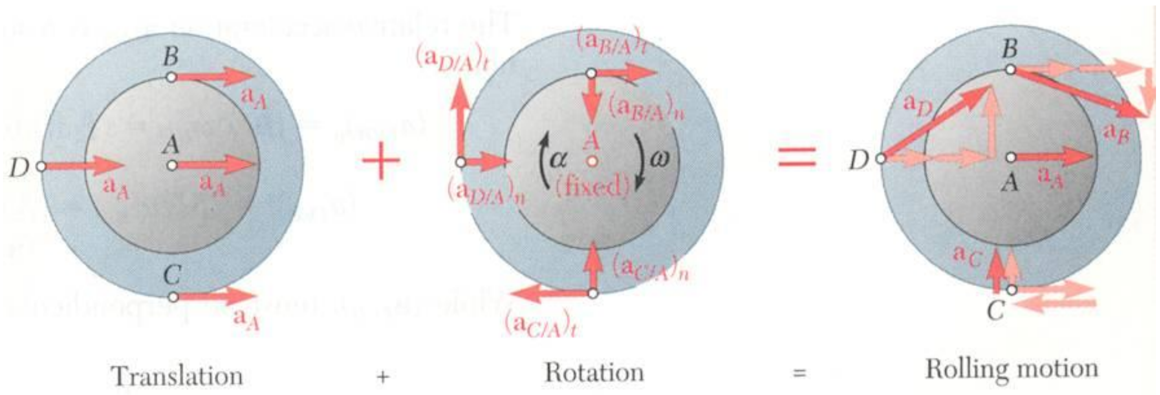
$$\omega = -\frac{v_A}{r_1} = -\frac{1.2\,\mathrm{m/s}}{0.150\,\mathrm{m}} = -8\,\mathrm{rad/s}$$

$$a_A = -r_1\ddot{\theta} = -r_1\alpha$$

$$\alpha = -\frac{a_A}{r_1} = -\frac{3\,\mathrm{m/s^2}}{0.150\,\mathrm{m}}$$

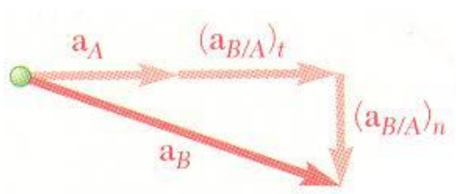
$$\boxed{\vec{\alpha} = \alpha\vec{k} = -\left(20\,\mathrm{rad/s^2}\right)\vec{k}}$$

Sample Problem 4



- The acceleration of each point is obtained by adding the acceleration of the gear center and the relative accelerations with respect to the center.

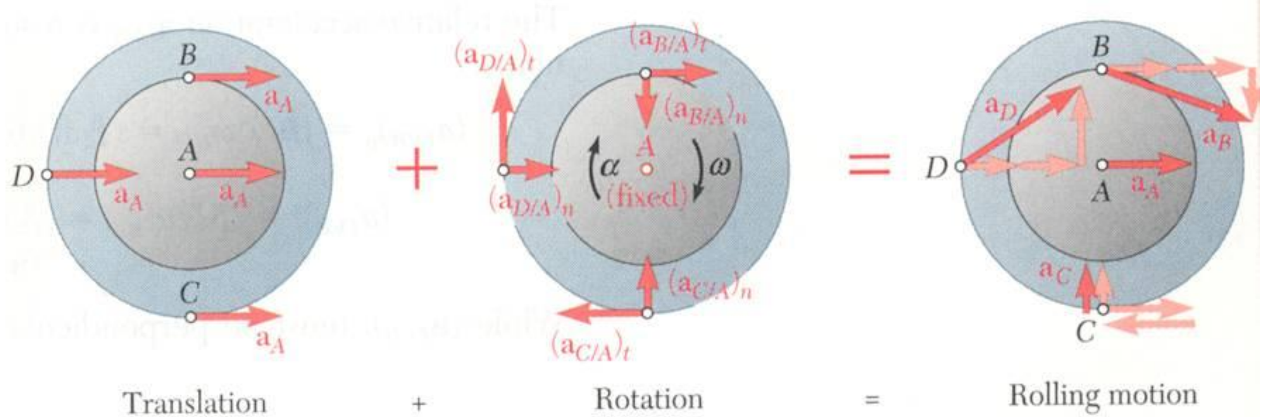
  The latter includes normal and tangential acceleration components.

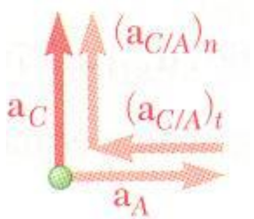


$$
\begin{aligned}
\vec{a}_B &= \vec{a}_A + \vec{a}_{B/A} = \vec{a}_A + \left(\vec{a}_{B/A}\right)_t + \left(\vec{a}_{B/A}\right)_n \\
&= \vec{a}_A + \alpha\vec{k} \times \vec{r}_{B/A} - \omega^2 \vec{r}_{B/A} \\
&= (3\,\mathrm{m/s^2})\vec{\imath} - (20\,\mathrm{rad/s^2})\vec{k} \times (0.100\mathrm{m})\vec{\jmath} \\
&\quad - (8\,\mathrm{rad/s})^2(0.100\mathrm{m})\vec{\jmath} \\
&= (3\,\mathrm{m/s^2})\vec{\imath} + (2\,\mathrm{m/s^2})\vec{\imath} - (6.40\,\mathrm{m/s^2})\vec{\jmath}
\end{aligned}
$$

$$\boxed{\vec{a}_B = \left(5\,\mathrm{m}/s^2\right)\vec{i} - \left(6.40\,\mathrm{m}/\mathrm{s}^2\right)\vec{j} \qquad a_B = 8.12\,\mathrm{m}/\mathrm{s}^2}$$

Sample Problem 4



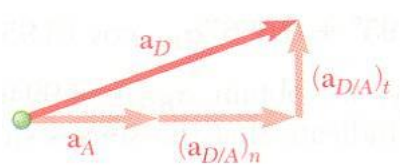


$$\vec{a}_C = \vec{a}_A + \vec{a}_{C/A} = \vec{a}_A + \alpha\vec{k}\times\vec{r}_{C/A} - \omega^2\vec{r}_{C/A}$$

$$= \left(3\,\mathrm{m/s^2}\right)\vec{i} - \left(20\,\mathrm{rad/s^2}\right)\vec{k}\times(-0.150\,\mathrm{m})\vec{j} - (8\,\mathrm{rad/s})^2(-0.150\,\mathrm{m})\vec{j}$$

$$= \left(3\,\mathrm{m/s^2}\right)\vec{i} - \left(3\,\mathrm{m/s^2}\right)\vec{i} + \left(9.60\,\mathrm{m/s^2}\right)\vec{j}$$

$$\boxed{\vec{a}_c = \left(9.60\,\mathrm{m/s^2}\right)\vec{j}}$$



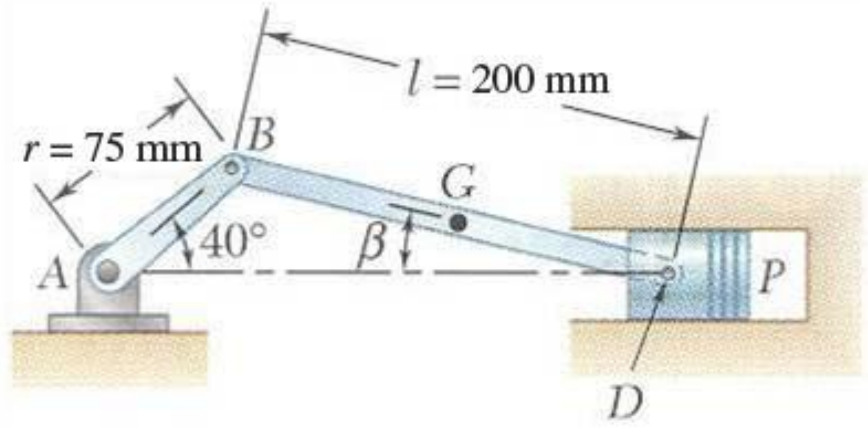
$$\vec{a}_D = \vec{a}_A + \vec{a}_{D/A} = \vec{a}_A + \alpha\vec{k}\times\vec{r}_{D/A} - \omega^2\vec{r}_{D/A}$$

$$= \left(3\,\mathrm{m/s^2}\right)\vec{i} - \left(20\,\mathrm{rad/s^2}\right)\vec{k}\times(-0.150\,\mathrm{m})\vec{i} - (8\,\mathrm{rad/s})^2(-0.150\mathrm{m})\vec{i}$$

$$= \left(3\,\mathrm{m/s^2}\right)\vec{i} + \left(3\,\mathrm{m/s^2}\right)\vec{j} + \left(9.60\,\mathrm{m/s^2}\right)\vec{i}$$

$$\boxed{\vec{a}_D = \left(12.6\,\mathrm{m/s^2}\right)\vec{i} + \left(3\,\mathrm{m/s^2}\right)\vec{j} \qquad a_D = 12.95\,\mathrm{m/s^2}}$$

Sample Problem 5



Crank *AB* of the engine system has a constant clockwise angular velocity of 2000 rpm.

For the crank position shown, determine the angular acceleration of the connecting rod *BD* and the acceleration of point *D*.

SOLUTION:

- The angular acceleration of the connecting rod *BD* and the acceleration of point *D* will be determined from

$$\vec{a}_D = \vec{a}_B + \vec{a}_{D/B} = \vec{a}_B + \left(\vec{a}_{D/B}\right)_t + \left(\vec{a}_{D/B}\right)_n$$

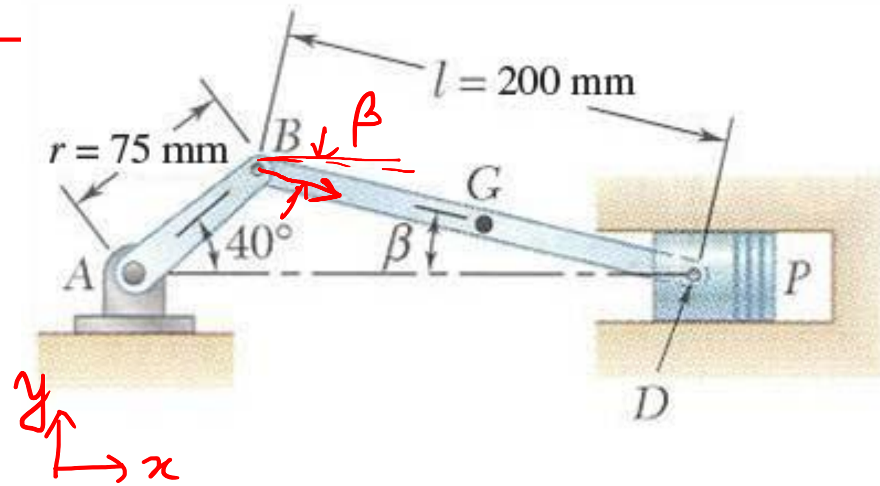
- The acceleration of *B* is determined from the given rotational speed of *AB*.
- The directions of the accelerations $\vec{a}_D, \left(\vec{a}_{D/B}\right)_t$, and $\left(\vec{a}_{D/B}\right)_n$ are determined from the geometry.
- Component equations for acceleration of point *D* are solved simultaneously for acceleration of *D* and angular acceleration of the connecting rod.

Sample Problem 5

$$\vec{\boldsymbol{\omega}}_{AB} = -2000\ \widehat{\boldsymbol{k}}\ \text{rpm}$$
$$= -\ 209.44\ \widehat{\boldsymbol{k}}\ \text{rad/s}$$

$$\vec{\boldsymbol{r}}_B = 0.075\ (\cos 40\ \hat{\boldsymbol{\imath}} + \ \sin 40\ \hat{\boldsymbol{\jmath}}\ )$$
$$= 0.0574\ \hat{\boldsymbol{\imath}} + 0.0482\ \hat{\boldsymbol{\jmath}}$$

$$\vec{\boldsymbol{v}}_B = \vec{\boldsymbol{\omega}}_{AB} \times \vec{\boldsymbol{r}}_B$$
$$= \ \text{-}12.033\hat{\boldsymbol{\jmath}} + 10.097\ \hat{\boldsymbol{\imath}}$$



From geometry,
$$0.075 \sin 40 = 0.2 \sin\beta \ \Rightarrow \beta = 13.95$$

$$\vec{\boldsymbol{v}}_D = \ v_D\hat{\boldsymbol{\imath}}$$
$$\vec{\boldsymbol{r}}_{D/B} = 0.2(\cos\beta\ \hat{\boldsymbol{\imath}}\ - \sin\beta\ \hat{\boldsymbol{\jmath}}\ )$$

$$\therefore \vec{\boldsymbol{r}}_{D/B} = 0.194\hat{\boldsymbol{\imath}}\ - 0.048\hat{\boldsymbol{\jmath}}$$

$$\vec{\boldsymbol{\omega}}_{\boldsymbol{DB}} = \ \omega_{DB}\widehat{\boldsymbol{k}}$$
$$\vec{\boldsymbol{v}}_D = \vec{\boldsymbol{v}}_B + \vec{\boldsymbol{\omega}}_{DB} \times \vec{\boldsymbol{r}}_{D/B} \ = \ \vec{\boldsymbol{v}}_B + \vec{\boldsymbol{v}}_{D/B}$$

$$v_D\hat{\boldsymbol{\imath}} = 10.097\hat{\boldsymbol{\imath}}\ - 12.033\hat{\boldsymbol{\jmath}} + \ \omega_{DB}\widehat{\boldsymbol{k}} \times (0.194\hat{\boldsymbol{\imath}}\ - 0.048\hat{\boldsymbol{\jmath}})$$

$$v_D\hat{\boldsymbol{\imath}} = 10.097\hat{\boldsymbol{\imath}}\ - 12.033\hat{\boldsymbol{\jmath}} + 0.194\ \omega_{DB}\hat{\boldsymbol{\jmath}} + 0.048\ \omega_{DB}\hat{\boldsymbol{\imath}}$$

From $\hat{\boldsymbol{\jmath}}$ component, $\omega_{DB} = 62.026$

Then from $\hat{\boldsymbol{\imath}}$ component, $v_D = 13.074$.

Sample Problem 5

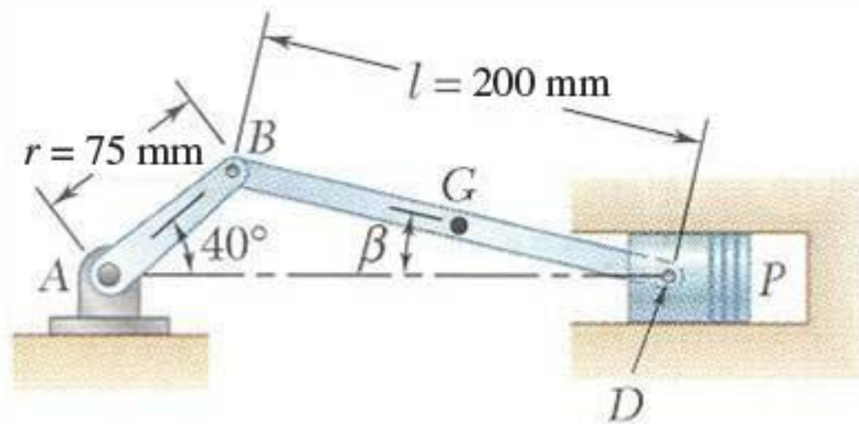
Acceleration :

$\vec{\boldsymbol{\alpha}}_{AB} = 0$ ($\because\ \omega_{AB}$ is constant)

$\vec{\boldsymbol{a}}_B = \vec{\boldsymbol{\alpha}}_{AB} \times \vec{\boldsymbol{r}}_B + \vec{\boldsymbol{\omega}}_{AB} \times \ \vec{\boldsymbol{\omega}}_{AB} \times \vec{\boldsymbol{r}}_B$

$= -209.44\,\widehat{\boldsymbol{k}} \times -209.44\,\widehat{\boldsymbol{k}} \times (0.0574\,\hat{\boldsymbol{\imath}} + 0.0482\,\hat{\boldsymbol{\jmath}})$

$= -2520.197\,\hat{\boldsymbol{\imath}} - 2114.7\,\hat{\boldsymbol{\jmath}}$



$\vec{\boldsymbol{a}}_D = \ a_D\,\hat{\boldsymbol{\imath}}$

$\vec{\boldsymbol{a}}_D = \vec{\boldsymbol{a}}_B + (\vec{\boldsymbol{a}}_{D/B})_t + (\vec{\boldsymbol{a}}_{D/B})_n$

$= \ \vec{\boldsymbol{a}}_B + \alpha_{BD}\,\widehat{\boldsymbol{k}} \times \vec{\boldsymbol{r}}_{D/B} + \omega_{BD}\widehat{\boldsymbol{k}} \times \omega_{BD}\widehat{\boldsymbol{k}} \times \vec{\boldsymbol{r}}_{D/B}$

$a_D\hat{\boldsymbol{\imath}} = -2520.197\,\hat{\boldsymbol{\imath}} - 2114.7\,\hat{\boldsymbol{\jmath}} + \ \alpha_{BD}\widehat{\boldsymbol{k}} \times (0.194\hat{\boldsymbol{\imath}} - 0.048\hat{\boldsymbol{\jmath}}) - \omega_{BD}^2(0.194\hat{\boldsymbol{\imath}} - 0.048\hat{\boldsymbol{\jmath}})$

$a_D\hat{\boldsymbol{\imath}} = -2520.197\,\hat{\boldsymbol{\imath}} \ - 2114.7\,\hat{\boldsymbol{\jmath}} + 0.194\alpha_{BD}\hat{\boldsymbol{\jmath}} + 0.048\alpha_{BD}\hat{\boldsymbol{\imath}} \ - 746.75\hat{\boldsymbol{\imath}} + 185.494\hat{\boldsymbol{\jmath}}$

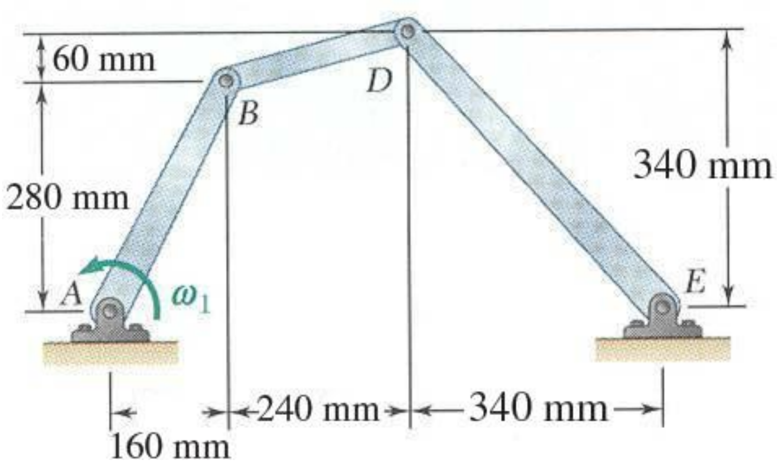
From $\hat{\boldsymbol{\jmath}}$ component,
α_{BD} = 9944.36 rad/s^2

From $\hat{\boldsymbol{\imath}}$ component,
$a_D \ = -2789.62$

$\therefore\ \vec{\boldsymbol{\alpha}}_{BD} \ = 9944.36\,\widehat{\boldsymbol{k}}$ rad/s^2

$\vec{\boldsymbol{a}}_D \ = -2789.62\,\hat{\boldsymbol{\imath}}$ m/s^2

Sample Problem 6



In the position shown, crank *AB* has a constant angular velocity ω_1 = 20 rad/s counterclockwise.

Determine the angular velocities and angular accelerations of the connecting rod *BD* and crank *DE*.

SOLUTION:

- The angular velocities are determined by simultaneously solving the component equations for

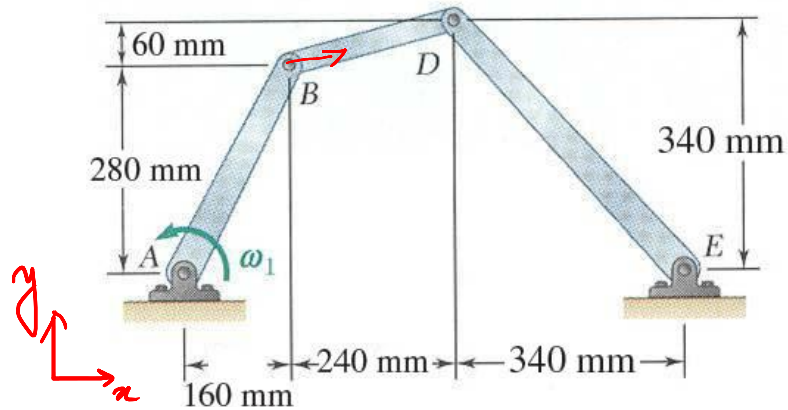
$$\vec{v}_D = \vec{v}_B + \vec{v}_{D/B}$$

- The angular accelerations are determined by simultaneously solving the component equations for

$$\vec{a}_D = \vec{a}_B + \vec{a}_{D/B}$$

Sample Problem 6

$\vec{\boldsymbol{\omega}}_{AB} = 20\widehat{\boldsymbol{k}}$
$\vec{\boldsymbol{r}}_B = 0.16\,\hat{\boldsymbol{\imath}} + 0.28\,\widehat{\boldsymbol{k}}$
$\vec{\boldsymbol{\alpha}}_{AB} = 0$



$$\vec{\boldsymbol{v}}_B = \vec{\boldsymbol{\omega}}_{AB} \times \vec{\boldsymbol{r}}_B$$
$$= 20\,\widehat{\boldsymbol{k}} \times (\,0.16\hat{\boldsymbol{\imath}} + 0.28\hat{\boldsymbol{\jmath}}\,)$$
$$= 3.2\,\hat{\boldsymbol{\jmath}} - 5.6\,\hat{\boldsymbol{\imath}}$$

$$\vec{\boldsymbol{a}}_B = \vec{\boldsymbol{\alpha}}_{AB} \times \vec{\boldsymbol{r}}_B + \vec{\boldsymbol{\omega}}_{AB} \times \vec{\boldsymbol{\omega}}_{AB} \times \vec{\boldsymbol{r}}_B$$
$$\vec{\boldsymbol{a}}_B = -20^2\,(0.16\,\hat{\boldsymbol{\imath}} + 0.28\,\hat{\boldsymbol{\jmath}}\,)$$
$$= -64\,\hat{\boldsymbol{\imath}} - 112\,\hat{\boldsymbol{\jmath}}$$

$$\vec{\boldsymbol{v}}_D = \omega_{DE}\widehat{\boldsymbol{k}} \times \vec{\boldsymbol{r}}_{D/E}$$
$$= \omega_{DE}\widehat{\boldsymbol{k}} \times (\,-0.34\,\hat{\boldsymbol{\imath}} + 0.34\hat{\boldsymbol{\jmath}}\,)$$
$$= -0.34\,\omega_{DE}\hat{\boldsymbol{\jmath}} - 0.34\omega_{DE}\hat{\boldsymbol{\imath}}$$

$$\vec{\boldsymbol{v}}_D = \vec{\boldsymbol{v}}_B + \vec{\boldsymbol{v}}_{D/B} = \vec{\boldsymbol{v}}_B + \omega_{BD}\widehat{\boldsymbol{k}} \times \vec{\boldsymbol{r}}_{D/B} = \vec{\boldsymbol{v}}_B + \omega_{BD}\widehat{\boldsymbol{k}} \times (0.24\hat{\boldsymbol{\imath}} + 0.06\hat{\boldsymbol{\jmath}}\,)$$
$$-0.34\,\omega_{DE}\hat{\boldsymbol{\jmath}} - 0.34\omega_{DE}\hat{\boldsymbol{\imath}} = 3.2\,\hat{\boldsymbol{\jmath}} - 5.6\hat{\boldsymbol{\imath}} + 0.24\omega_{BD}\hat{\boldsymbol{\jmath}} - 0.06\omega_{BD}\hat{\boldsymbol{\imath}}$$

Equating $\hat{\boldsymbol{\imath}}$ & $\hat{\boldsymbol{\jmath}}$ components,
$$0.34\,\omega_{DE} - 0.06\omega_{BD} = 5.6$$
$$0.34\,\omega_{DE} + 0.24\omega_{BD} = -3.2$$

Solving we get, $\omega_{BD} = -29.333$
$\omega_{DE} = 11.294$

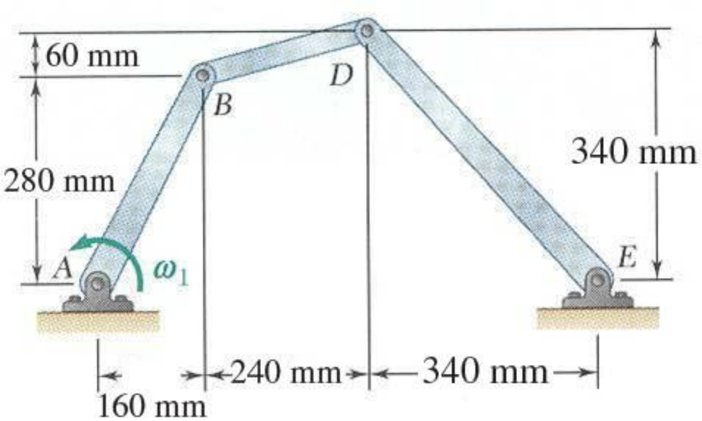
$$\therefore \vec{\boldsymbol{\omega}}_{BD} = -29.333\widehat{\boldsymbol{k}}\text{ rad/s}$$
$$\vec{\boldsymbol{\omega}}_{DE} = 11.294\,\widehat{\boldsymbol{k}}\text{ rad/s}$$

Sample Problem 6

$\vec{\boldsymbol{a}}_D = \alpha_{DE}\widehat{\boldsymbol{k}} \times \vec{\boldsymbol{r}}_{D/E} + \omega_{DE}\widehat{\boldsymbol{k}} \times \omega_{DE}\widehat{\boldsymbol{k}} \times \vec{\boldsymbol{r}}_{D/E}$

$= \alpha_{DE}\widehat{\boldsymbol{k}} \times (-0.34\,\hat{\boldsymbol{\imath}} + 0.34\hat{\boldsymbol{\jmath}}) - \omega_{DE}^2(-0.34\,\hat{\boldsymbol{\imath}} + 0.34\hat{\boldsymbol{\jmath}})$

$= -0.34\alpha_{DE}\hat{\boldsymbol{\jmath}} - 0.34\alpha_{DE}\hat{\boldsymbol{\imath}} + 43.368\hat{\boldsymbol{\imath}} - 43.368\hat{\boldsymbol{\jmath}}$



$\vec{\boldsymbol{a}}_{D/B} = \alpha_{BD}\widehat{\boldsymbol{k}} \times \vec{\boldsymbol{r}}_{D/B} + \omega_{DB}\widehat{\boldsymbol{k}} \times \omega_{DB}\widehat{\boldsymbol{k}} \times \vec{\boldsymbol{r}}_{D/B}$

$= \alpha_{BD}\widehat{\boldsymbol{k}} \times (0.24\hat{\boldsymbol{\imath}} + 0.06\hat{\boldsymbol{\jmath}}) - \omega_{BD}^2(0.24\hat{\boldsymbol{\imath}} + 0.06\hat{\boldsymbol{\jmath}})$

$= 0.24\,\alpha_{BD}\hat{\boldsymbol{\jmath}} - 0.06\alpha_{BD}\hat{\boldsymbol{\imath}} - 206.51\hat{\boldsymbol{\imath}} - 51.625\hat{\boldsymbol{\jmath}}$

$\vec{a}_D = \vec{a}_B + \vec{a}_{D/B}$

$-0.34\alpha_{DE}\hat{\boldsymbol{\jmath}} - 0.34\alpha_{DE}\hat{\boldsymbol{\imath}} + 43.368\hat{\boldsymbol{\imath}} - 43.368\hat{\boldsymbol{\jmath}} = -64\,\hat{\boldsymbol{\imath}} - 112\,\hat{\boldsymbol{\jmath}} + 0.24\,\alpha_{BD}\hat{\boldsymbol{\jmath}} - 0.06\alpha_{BD}\hat{\boldsymbol{\imath}} - 206.51\hat{\boldsymbol{\imath}} - 51.625\hat{\boldsymbol{\jmath}}$

Equating $\hat{\boldsymbol{\imath}}$ components

$0.06\alpha_{BD} - 0.34\alpha_{DE} = -313.878$

Equating $\hat{\boldsymbol{\jmath}}$ components

$0.24\alpha_{BD} + 0.34\alpha_{DE} = 120.2585$

Solving we get,

$\alpha_{BD} = -645.398$

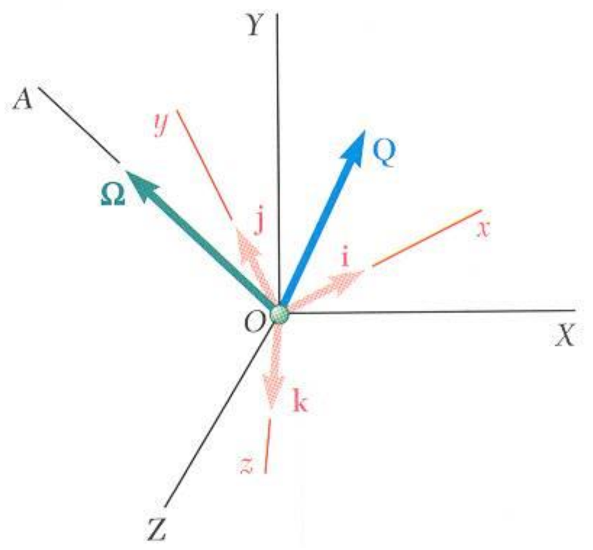
$\alpha_{DE} = 809.276$

∴Angular accelerations are

$\vec{\boldsymbol{\alpha}}_{DB} = -645.398\,\hat{k}$ rad/s^2

$\vec{\boldsymbol{\alpha}}_{DE} = 809.276\,\hat{k}$ rad/s^2

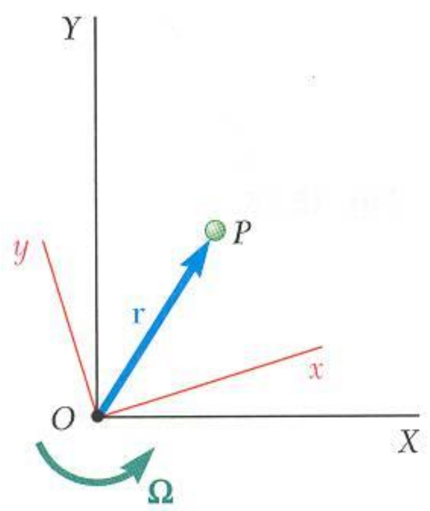
Rate of Change With Respect to a Rotating Frame



- Frame *OXYZ* is fixed.
- Frame *Oxyz* rotates about fixed axis *OA* with angular velocity $\vec{\Omega}$
- Vector function $\vec{Q}(t)$ varies in direction and magnitude.

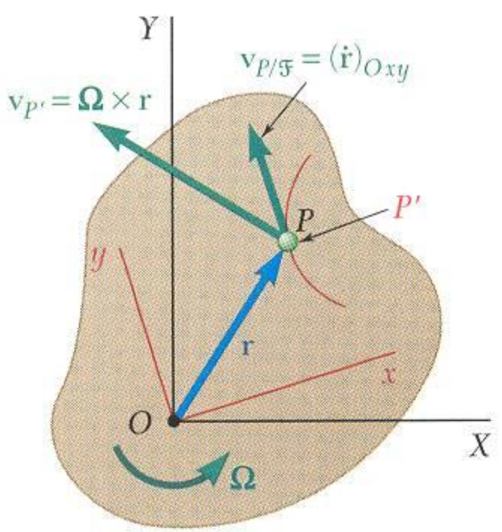
- With respect to the rotating *Oxyz* frame,
$$\vec{Q} = Q_x\vec{i} + Q_y\vec{j} + Q_z\vec{k}$$
$$\left(\dot{\vec{Q}}\right)_{Oxyz} = \dot{Q}_x\vec{i} + \dot{Q}_y\vec{j} + \dot{Q}_z\vec{k}$$
- With respect to the fixed *OXYZ* frame,
$$\left(\dot{\vec{Q}}\right)_{OXYZ} = \dot{Q}_x\vec{i} + \dot{Q}_y\vec{j} + \dot{Q}_z\vec{k} + Q_x\dot{\vec{i}} + Q_y\dot{\vec{j}} + Q_z\dot{\vec{k}}$$
- $\dot{Q}_x\vec{i} + \dot{Q}_y\vec{j} + \dot{Q}_z\vec{k} = \left(\dot{\vec{Q}}\right)_{Oxyz}$ = rate of change with respect to rotating frame.
- If $\vec{Q}$ were fixed within *Oxyz* then $\left(\dot{\vec{Q}}\right)_{OXYZ}$ is equivalent to velocity of a point in a rigid body attached to *Oxyz* and $Q_x\dot{\vec{i}} + Q_y\dot{\vec{j}} + Q_z\dot{\vec{k}} = \vec{\Omega}\times\vec{Q}$
- With respect to the fixed *OXYZ* frame,
$$\left(\dot{\vec{Q}}\right)_{OXYZ} = \left(\dot{\vec{Q}}\right)_{Oxyz} + \vec{\Omega}\times\vec{Q}$$

Coriolis Acceleration



- Frame *OXY* is fixed and frame *Oxy* rotates with angular velocity $\vec{\Omega}$.
- Position vector $\vec{r}_P$ for the particle *P* is the same in both frames but the rate of change depends on the choice of frame.
- The absolute velocity of the particle *P* is

$$\vec{v}_P = \left(\dot{\vec{r}}\right)_{OXY} = \vec{\Omega} \times \vec{r} + \left(\dot{r}\right)_{Oxy}$$



- Imagine a rigid slab attached to the rotating frame *Oxy* or $\mathfrak{F}$ for short. Let *P′* be a point on the slab which corresponds instantaneously to position of particle *P*.

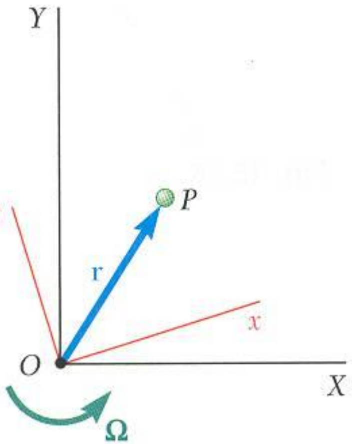
$$\vec{v}_{P/\mathbf{F}} = \left(\dot{\vec{r}}\right)_{Oxy} = \quad \text{velocity of } P \text{ along its path on the slab}$$

$$\vec{v}_{P'} = \text{velocity of point } P' \text{ on the slab}$$

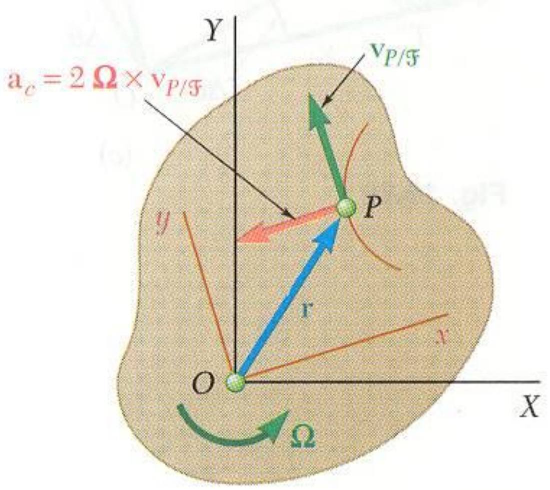
- Absolute velocity for the particle P may be written as

$$\vec{v}_P = \vec{v}_{P¢} + \vec{v}_{P/\mathbf{F}}$$

Coriolis Acceleration



$$\vec{v}_P = \vec{\Omega}\times\vec{r} + (\dot{r})_{Oxy}$$
$$= \vec{v}_{P'} + \vec{v}_{P/\mathfrak{F}}$$



- Absolute acceleration for the particle P is

$$\vec{a}_P = \dot{\vec{\Omega}}\times\vec{r} + \vec{\Omega}\times(\dot{\vec{r}})_{OXY} + \frac{d}{dt}\left[(\dot{\vec{r}})_{Oxy}\right]$$

but, $\quad (\dot{\vec{r}})_{OXY} = \vec{\Omega}\times\vec{r} + (\dot{\vec{r}})_{Oxy}$

$$\frac{d}{dt}\left[(\dot{\vec{r}})_{Oxy}\right] = (\ddot{\vec{r}})_{Oxy} + \vec{\Omega}\times(\dot{\vec{r}})_{Oxy}$$

$$\vec{a}_P = \dot{\vec{\Omega}}\times\vec{r} + \vec{\Omega}\times(\vec{\Omega}\times\vec{r}) + 2\vec{\Omega}\times(\dot{\vec{r}})_{Oxy} + (\ddot{\vec{r}})_{Oxy}$$

- Utilizing the conceptual point P' on the slab,

$$\vec{a}_{P'} = \dot{\vec{\Omega}}\times\vec{r} + \vec{\Omega}\times(\vec{\Omega}\times\vec{r})$$

$$\vec{a}_{P/\mathfrak{F}} = (\ddot{\vec{r}})_{Oxy}$$

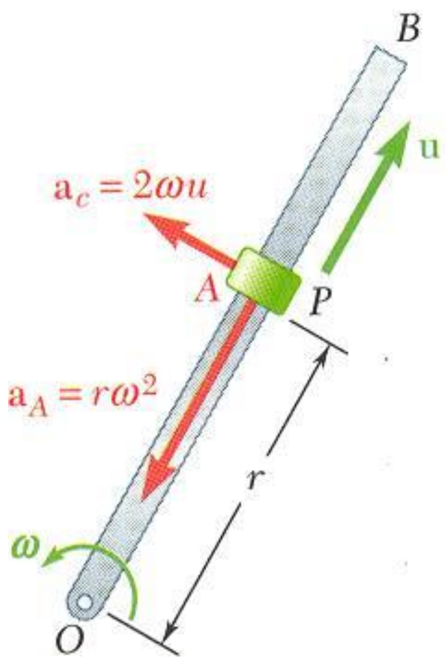
- Absolute acceleration for the particle P becomes

$$\vec{a}_P = \vec{a}_{P'} + \vec{a}_{P/\mathfrak{F}} + 2\vec{\Omega}\times(\dot{\vec{r}})_{Oxy}$$

$$= \vec{a}_{P'} + \vec{a}_{P/\mathfrak{F}} + \vec{a}_c$$

$$\vec{a}_c = 2\vec{\Omega}\times(\dot{\vec{r}})_{Oxy} = 2\vec{\Omega}\times\vec{v}_{P/\mathfrak{F}} = \text{Coriolis acceleration}$$

Coriolis Acceleration



- Consider a collar *P* which is made to slide at constant relative velocity *u* along rod *OB*. The rod is rotating at a constant angular velocity *ω*. The point *A* on the rod corresponds to the instantaneous position of *P*.
- Absolute acceleration of the collar is

$$\vec{a}_P = \vec{a}_A + \vec{a}_{P/\mathbf{F}} + \vec{a}_c$$

where

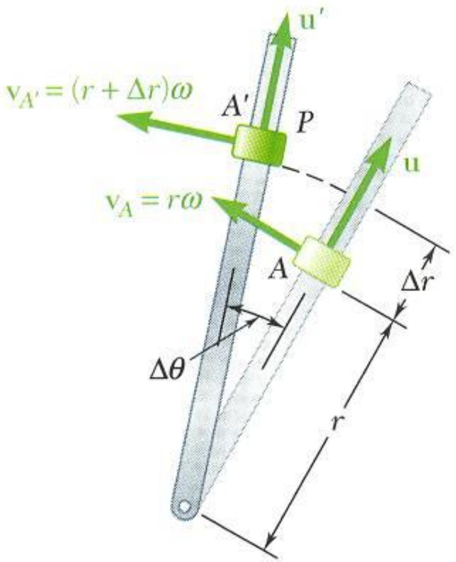
$$\vec{a}_A = \dot{\vec{\Omega}} \times \vec{r} + \vec{\Omega} \times \left(\vec{\Omega} \times \vec{r}\right) \qquad a_A = r\omega^2$$

$$\vec{a}_{P/\mathbf{F}} = \left(\ddot{\vec{r}}\right)_{Oxy} = 0$$

$$\vec{a}_c = 2\vec{\Omega} \times \vec{v}_{P/\mathbf{F}} \qquad a_c = 2\omega u$$

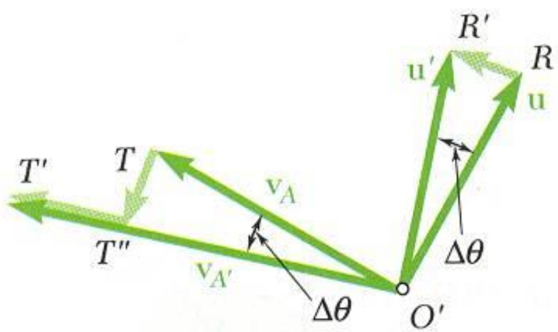
- The absolute acceleration consists of the radial and tangential vectors shown

Coriolis Acceleration



at t, $\qquad \vec{v} = \vec{v}_A + \vec{u}$

at $t + \Delta t$, $\qquad \vec{v}\,' = \vec{v}_{A'} + \vec{u}\,'$



- Change in velocity over Δt is represented by the sum of three vectors

$$\Delta\vec{v} = \overrightarrow{RR'} + \overrightarrow{TT''} + \overrightarrow{T''T'}$$

- $\overrightarrow{TT''}$ is due to change in direction of the velocity of point A on the rod,

$$\lim_{\Delta t \to 0} \frac{\overrightarrow{TT''}}{\Delta t} = \lim_{\Delta t \to 0} v_A \frac{\Delta\theta}{\Delta t} = r\omega\omega = r\omega^2 = a_A$$

$$\text{recall,} \quad \vec{a}_A = \dot{\vec{\Omega}} \times \vec{r} + \vec{\Omega} \times \left(\vec{\Omega} \times \vec{r}\right) \qquad a_A = r\omega^2$$

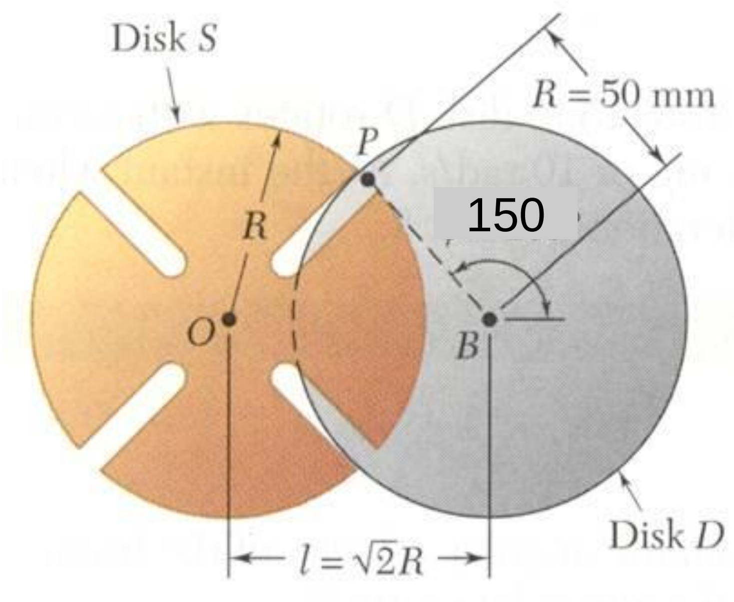
- $\overrightarrow{RR'}$ and $\overrightarrow{T''T'}$ result from combined effects of relative motion of P and rotation of the rod

$$\lim_{\Delta t \to 0}\left(\frac{\overrightarrow{RR'}}{\Delta t} + \frac{\overrightarrow{T''T'}}{\Delta t}\right) = \lim_{\Delta t \to 0}\left(u\frac{\Delta\theta}{\Delta t} + \omega\frac{\Delta r}{\Delta t}\right)$$

$$= u\omega + \omega u = 2\omega u$$

$$\text{recall,} \quad \vec{a}_c = 2\vec{\Omega} \times \vec{v}_{P/\mathbf{F}} \qquad a_c = 2\omega u$$

Sample Problem 7



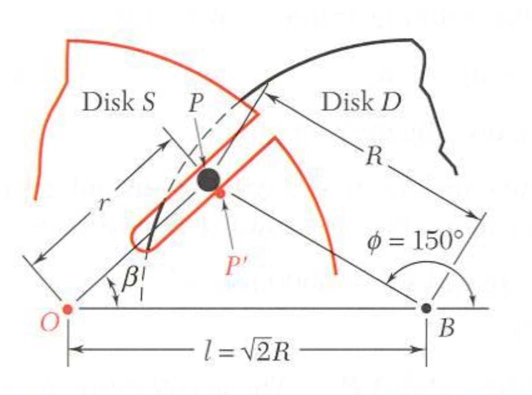
Disk D of the Geneva mechanism rotates with constant counterclockwise angular velocity ω_D = 10 rad/s.

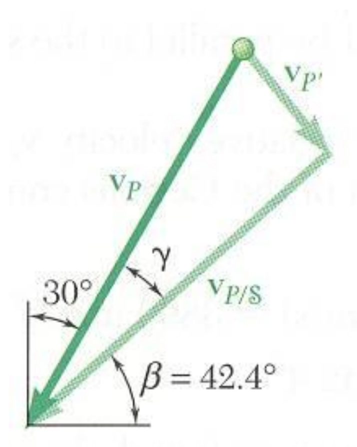
At the instant when ϕ = 150^o, determine (*a*) the angular velocity of disk *S*, and (*b*) the velocity of pin *P* relative to disk *S*.

SOLUTION:

- The absolute velocity of the point *P* may be written as
$$\vec{v}_P = \vec{v}_{P'} + \vec{v}_{P/s}$$
- Magnitude and direction of velocity $\vec{v}_P$ of pin *P* are calculated from the radius and angular velocity of disk *D*.
- Direction of velocity $\vec{v}_{P'}$ of point *P*' on *S* coinciding with *P* is perpendicular to radius *OP*.
- Direction of velocity $\vec{v}_{P/s}$ of *P* with respect to *S* is parallel to the slot.
- Solve the vector triangle for the angular velocity of *S* and relative velocity of *P*.

Sample Problem 7





SOLUTION:

- The absolute velocity of the point P may be written as

$$\vec{v}_P = \vec{v}_{P'} + \vec{v}_{P/s}$$

- Magnitude and direction of absolute velocity of pin P are calculated from radius and angular velocity of disk D.

$$v_P = R\omega_D = (50\,\text{mm})(10\,\text{rad/s}) = 500\,\text{mm/s}$$

- Direction of velocity of P with respect to S is parallel to slot. From the law of cosines,

$$r^2 = R^2 + l^2 - 2Rl\cos 30° = 0.551R^2 \qquad r = 37.1\,\text{mm}$$

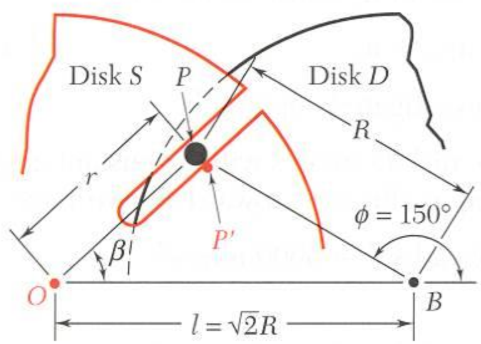
From the law of cosines,

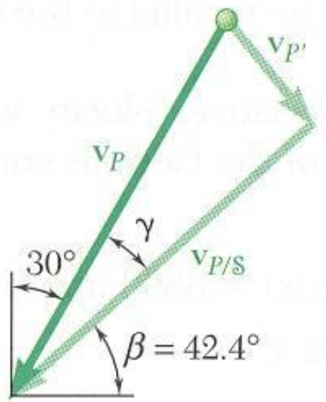
$$\frac{\sin\beta}{\text{R}} = \frac{\sin 30°}{r} \qquad \sin\beta = \frac{\sin 30°}{0.742} \qquad \beta = 42.4°$$

The interior angle of the vector triangle is

$$\gamma = 90° - 42.4° - 30° = 17.6°$$

Sample Problem 7





$$v_P = 500\,\text{mm/s}$$

- Direction of velocity of point P' on S coinciding with P is perpendicular to radius OP. From the velocity triangle,

$$v_{P'} = v_P \sin\gamma = (500\,\text{mm/s})\sin 17.6° = 151.2\,\text{mm/s}$$

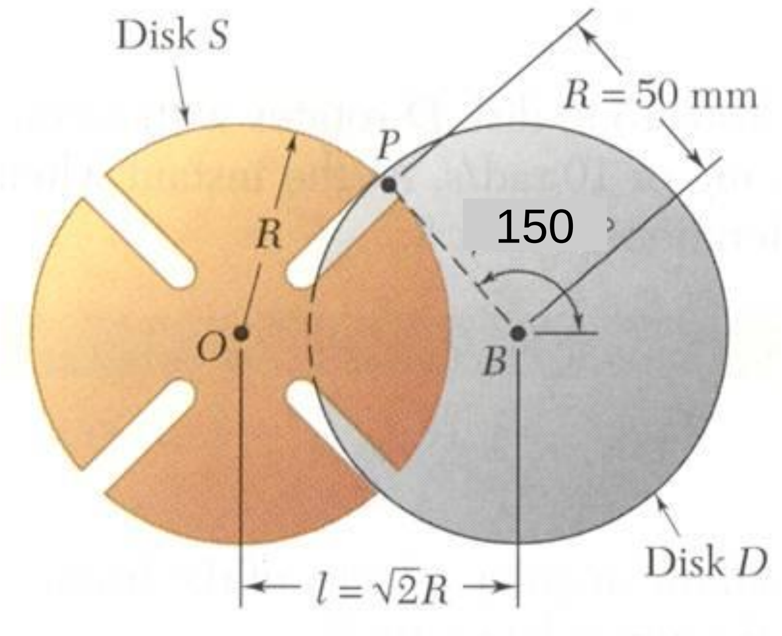
$$= r\omega_s \qquad \omega_s = \frac{151.2\,\text{mm/s}}{37.1\,\text{mm}}$$

$$\boxed{\vec{\omega}_s = (-4.08\,\text{rad/s})\vec{k}}$$

$$v_{P/s} = v_P \cos\gamma = (500\,\text{m/s})\cos 17.6°$$

$$\boxed{\vec{v}_{P/s} = (477\,\text{m/s})\left(-\cos 42.4°\,\vec{i} - \sin 42.4°\,\vec{j}\right)}$$

Sample Problem 8



In the Geneva mechanism, disk D rotates with a constant counter-clockwise angular velocity of 10 rad/s. At the instant when $\varphi = 150^o$, determine angular acceleration of disk S.

SOLUTION:

- The absolute acceleration of the pin P may be expressed as

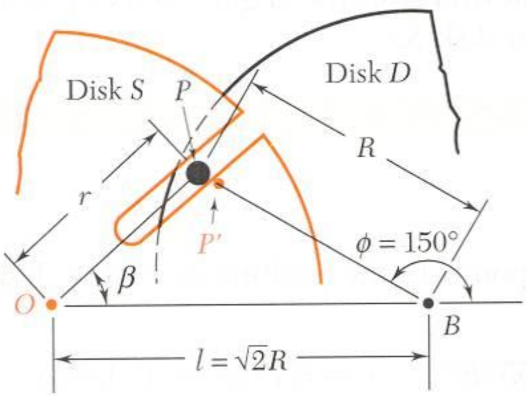
$$\vec{a}_P = \vec{a}_{P'} + \vec{a}_{P/s} + \vec{a}_c$$

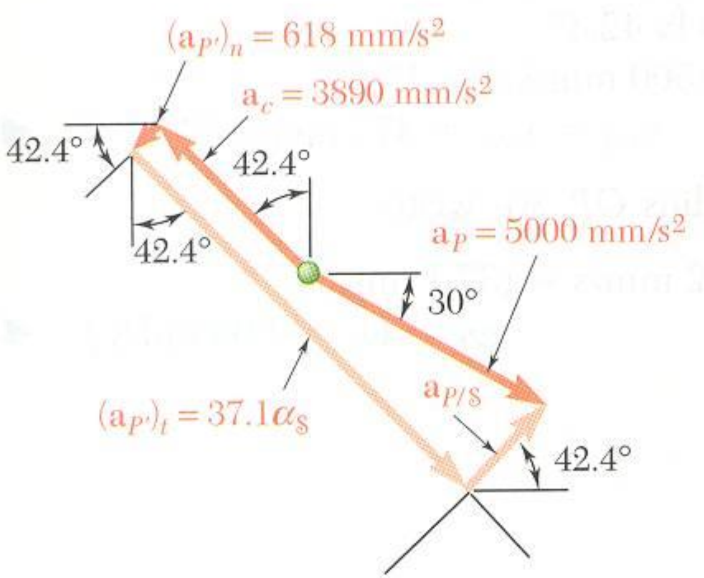
- The instantaneous angular velocity of Disk S is determined as in previous Problem.

- The only unknown involved in the acceleration equation is the instantaneous angular acceleration of Disk S.

- Resolve each acceleration term into the component parallel to the slot. Solve for the angular acceleration of Disk S.

Sample Problem 8





SOLUTION:

- Absolute acceleration of the pin P may be expressed as

$$\vec{a}_P = \vec{a}_{P'} + \vec{a}_{P/s} + \vec{a}_c$$

- From previous Problem.

$$\beta = 42.4° \qquad \vec{\omega}_S = (-4.08\,\text{rad/s})\vec{k}$$

$$\vec{v}_{P/s} = (477\,\text{mm/s})(-\cos 42.4°\vec{i} - \sin 42.4°\vec{j})$$

- Considering each term in the acceleration equation,

$$a_P = R\omega_D{}^2 = 5000\ \text{mm/s}^2$$
$$\overrightarrow{a_P} = 5000(\text{Cos}30\vec{\imath} - \text{Sin}30\vec{\jmath})\text{mm/s}^2$$

$$\vec{a}_{P'} = (\vec{a}_{P'})_n + (\vec{a}_{P'})_t$$

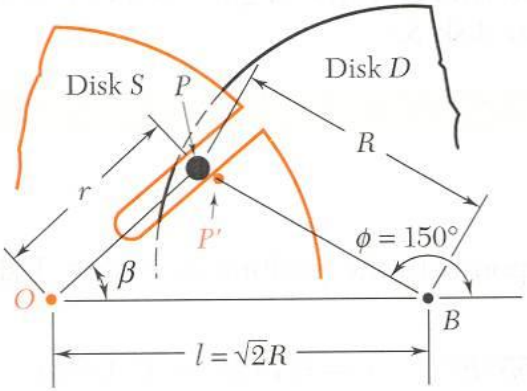
$$(\vec{a}_{P'})_n = (r\omega_S^2)(-\cos 42.4°\vec{i} - \sin 42.4°\vec{j})$$

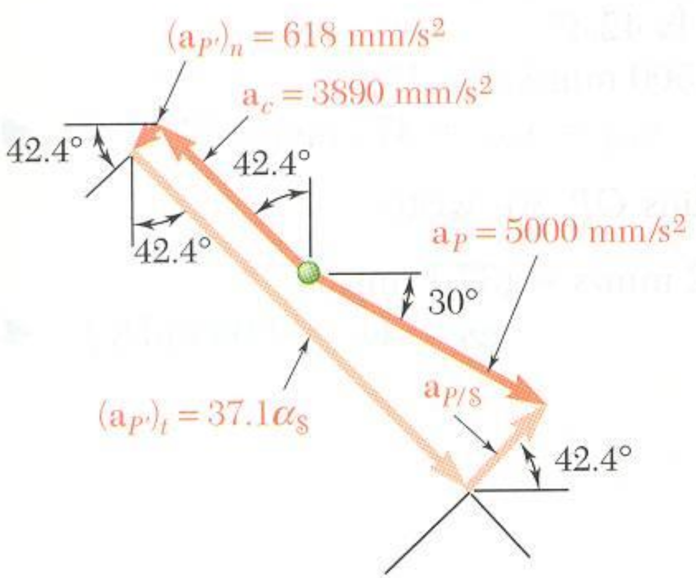
$$(\vec{a}_{P'})_t = (r\alpha_S)(-\sin 42.4°\vec{i} + \cos 42.4°\vec{j})$$

$$(\vec{a}_{P'})_t = (\alpha_S)(37.1\text{mm})(-\sin 42.4°\vec{i} + \cos 42.4°\vec{j})$$

note: α_S may be positive or negative

Sample Problem 8





- The direction of the Coriolis acceleration is obtained by rotating the direction of the relative velocity $\vec{v}_{P/s}$ by 90° in the sense of ω_S.

$$\vec{a}_c = (2\omega_S v_{P/s})(-\sin 42.4°\vec{i} + \cos 42.4\vec{j})$$
$$= 2(4.08\,\text{rad/s})(477\,\text{mm/s})(-\sin 42.4°\vec{i} + \cos 42.4\vec{j})$$
$$= (3890\,\text{mm/s}^2)(-\sin 42.4°\vec{i} + \cos 42.4\vec{j})$$

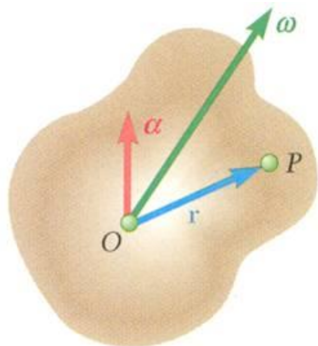
- The relative acceleration $\vec{a}_{P/s}$ must be parallel to the slot.

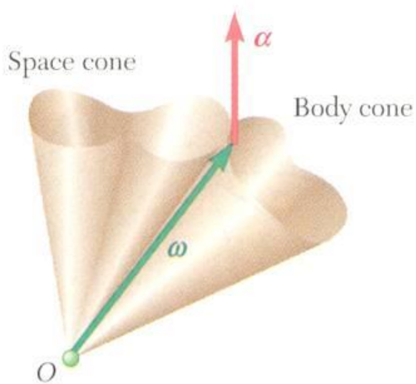
- Equating components of the acceleration terms perpendicular to the slot,

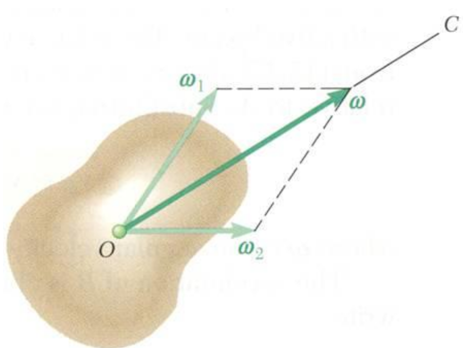
$$37.1\alpha_s - 3890 = 5000\text{Cos } 17.7$$
$$\alpha_s = 233$$

$$\overrightarrow{\boldsymbol{\alpha_s}} = -233\ \text{rad/s}^2\overrightarrow{\boldsymbol{k}}$$

Motion About a Fixed Point







- The most general displacement of a rigid body with a fixed point *O* is equivalent to a rotation of the body about an axis through *O*.
- With the instantaneous axis of rotation and angular velocity $\vec{\omega}$, the velocity of a particle *P* of the body is

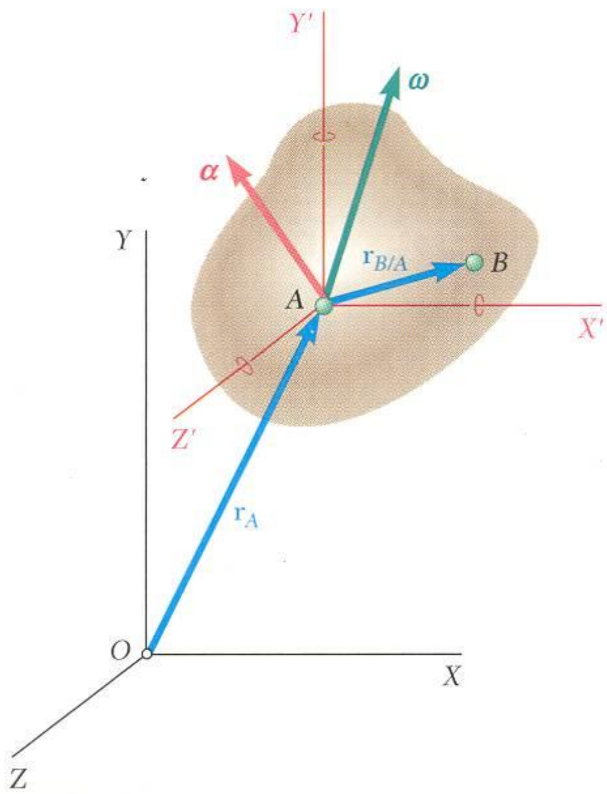
$$\vec{v} = \frac{d\vec{r}}{dt} = \vec{\omega} \times \vec{r}$$

and the acceleration of the particle *P* is

$$\vec{a} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \left(\vec{\omega} \times \vec{r}\right) \qquad \vec{\alpha} = \frac{d\vec{\omega}}{dt}.$$

- The angular acceleration $\vec{\alpha}$ represents the velocity of the tip of $\vec{\omega}$.
- As the vector $\vec{\omega}$ moves within the body and in space, it generates a body cone and space cone which are tangent along the instantaneous axis of rotation.
- Angular velocities have magnitude and direction and obey parallelogram law of addition. They are vectors.

General Motion



- For particles *A* and *B* of a rigid body,

$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$$

- Particle *A* is fixed within the body and motion of the body relative to *AX'Y'Z'* is the motion of a body with a fixed point

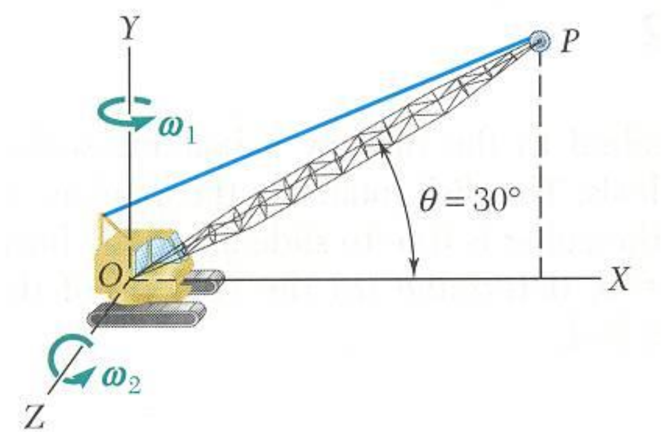
$$\vec{v}_B = \vec{v}_A + \vec{\omega} \times \vec{r}_{B/A}$$

- Similarly, the acceleration of the particle *P* is

$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$$
$$= \vec{a}_A + \vec{\alpha} \times \vec{r}_{B/A} + \vec{\omega} \times \left(\vec{\omega} \times \vec{r}_{B/A}\right)$$

- Most general motion of a rigid body is equivalent to:
  - a translation in which all particles have the same velocity and acceleration of a reference particle *A*, and
  - of a motion in which particle *A* is assumed fixed.

Sample Problem 9



The crane rotates with a constant angular velocity ω_1 = 0.30 rad/s and the boom is being raised with a constant angular velocity ω_2 = 0.50 rad/s. The length of the boom is l = 12 m.

Determine:

- angular velocity of the boom,
- angular acceleration of the boom,
- velocity of the boom tip, and
- acceleration of the boom tip.

SOLUTION:

With $\vec{\omega}_1 = 0.30\vec{j} \quad \vec{\omega}_2 = 0.50\vec{k}$

$$\vec{r} = 12\left(\cos 30°\vec{i} + \sin 30°\vec{j}\right)$$
$$= 10.39\vec{i} + 6\vec{j}$$

- Angular velocity of the boom,

$$\vec{\omega} = \vec{\omega}_1 + \vec{\omega}_2$$

- Angular acceleration of the boom,

$$\vec{\alpha} = \dot{\vec{\omega}}_1 + \dot{\vec{\omega}}_2 = \dot{\vec{\omega}}_2 = \left(\dot{\vec{\omega}}_2\right)_{Oxyz} + \vec{\Omega} \times \vec{\omega}_2$$
$$= \vec{\omega}_1 \times \vec{\omega}_2$$

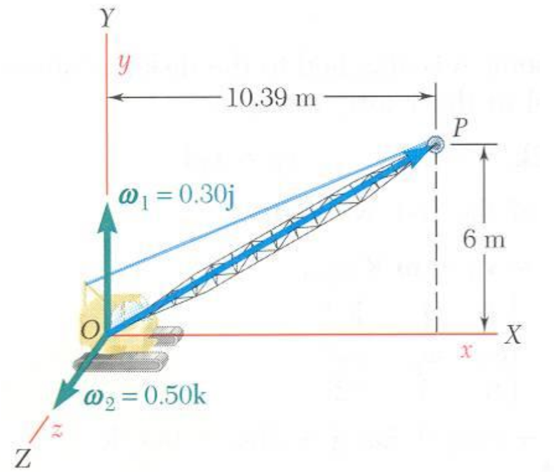
- Velocity of boom tip,

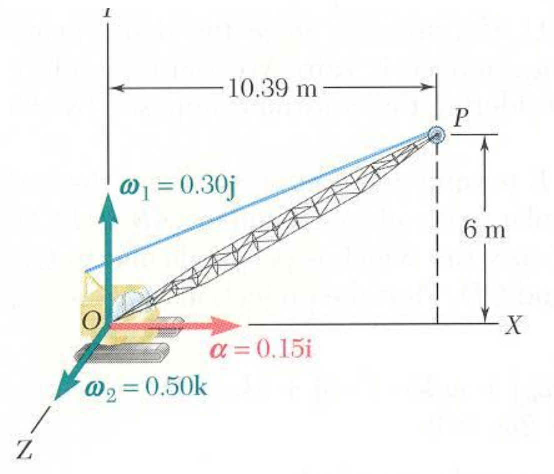
$$\vec{v} = \vec{\omega} \times \vec{r}$$

- Acceleration of boom tip,

$$\vec{a} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \left(\vec{\omega} \times \vec{r}\right) = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v}$$

Sample Problem 9





$$\vec{\omega}_1 = 0.30\vec{j} \quad \vec{\omega}_2 = 0.50\vec{k}$$

$$\vec{r} = 10.39\vec{i} + 6\vec{j}$$

<u>SOLUTION</u>:

- Angular velocity of the boom,

$$\vec{\omega} = \vec{\omega}_1 + \vec{\omega}_2$$

$$\boxed{\vec{\omega} = (0.30\,\mathrm{rad/s})\vec{j} + (0.50\,\mathrm{rad/s})\vec{k}}$$

- Angular acceleration of the boom,

$$\vec{\alpha} = \dot{\vec{\omega}}_1 + \dot{\vec{\omega}}_2 = \dot{\vec{\omega}}_2 = \left(\dot{\vec{\omega}}_2\right)_{Oxyz} + \vec{\Omega} \times \vec{\omega}_2$$

$$= \vec{\omega}_1 \times \vec{\omega}_2 = (0.30\,\mathrm{rad/s})\vec{j} \times (0.50\,\mathrm{rad/s})\vec{k}$$

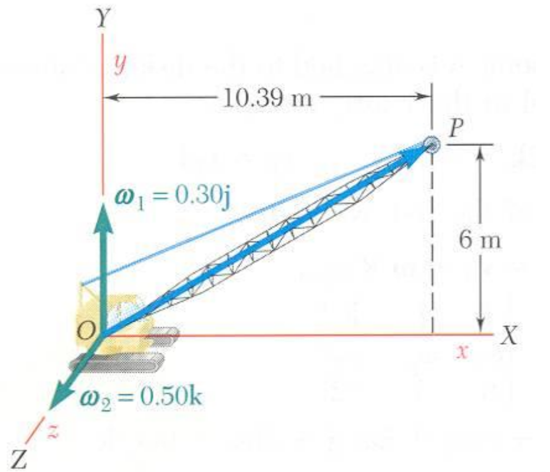
$$\boxed{\vec{\alpha} = \left(0.15\,\mathrm{rad/s^2}\right)\vec{i}}$$

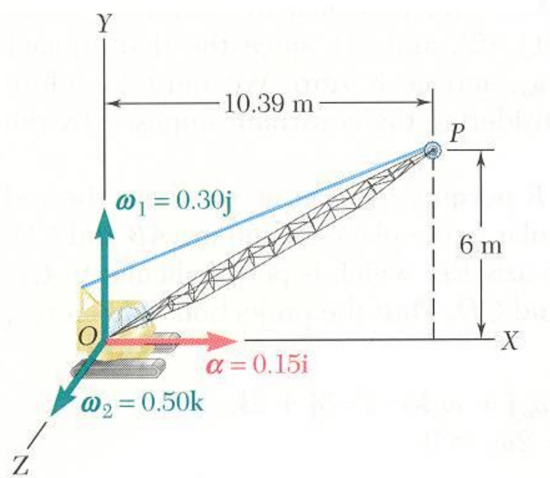
- Velocity of boom tip,

$$\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0.3 & 0.5 \\ 10.39 & 6 & 0 \end{vmatrix}$$

$$\boxed{\vec{\boldsymbol{v}} = -3\vec{\boldsymbol{\imath}} + 5.2\vec{\jmath} - 3.12\vec{\boldsymbol{k}}\ \mathrm{m/s}}$$

Sample Problem 9





$$\vec{\omega}_1 = 0.30\vec{j} \quad \vec{\omega}_2 = 0.50\vec{k}$$

$$\vec{r} = 10.39\vec{i} + 6\vec{j}$$

- Acceleration of boom tip,

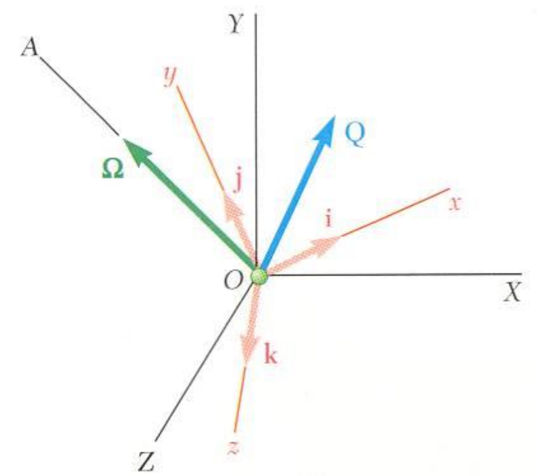
$$\vec{a} = \vec{\alpha}\times\vec{r} + \vec{\omega}\times\left(\vec{\omega}\times\vec{r}\right) = \vec{\alpha}\times\vec{r} + \vec{\omega}\times\vec{v}$$

$$\vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0.15 & 0 & 0 \\ 10.39 & 6 & 0 \end{vmatrix} + \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0.30 & 0.50 \\ -3 & 5.20 & -3.12 \end{vmatrix}$$

$$= 0.90\vec{k} - 0.94\vec{i} - 2.60\vec{i} - 1.50\vec{j} + 0.90\vec{k}$$

$$\boxed{\vec{a} = -\left(3.54\,\mathrm{m/s^2}\right)\vec{i} - \left(1.50\,\mathrm{m/s^2}\right)\vec{j} + \left(1.80\,\mathrm{m/s^2}\right)\vec{k}}$$

Three-Dimensional Motion. Coriolis Acceleration



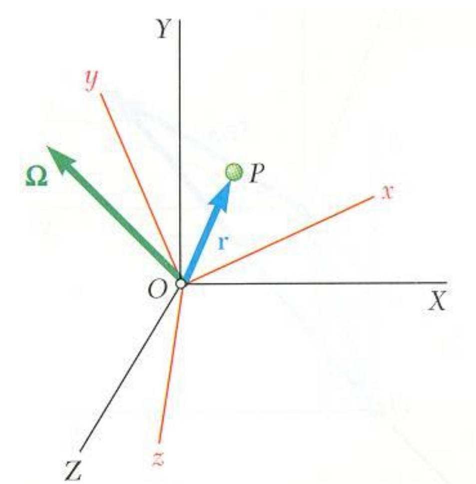
- With respect to the fixed frame *OXYZ* and rotating frame *Oxyz*,

$$\left(\dot{\vec{Q}}\right)_{OXYZ} = \left(\dot{\vec{Q}}\right)_{Oxyz} + \vec{\Omega} \times \vec{Q}$$

- Consider motion of particle *P* relative to a rotating frame *Oxyz* or $\mathscr{F}$ for short. The absolute velocity can be expressed as

$$\vec{v}_P = \vec{\Omega} \times \vec{r} + \left(\dot{\vec{r}}\right)_{Oxyz}$$

$$= \vec{v}_{P'} + \vec{v}_{P/\mathscr{F}}$$



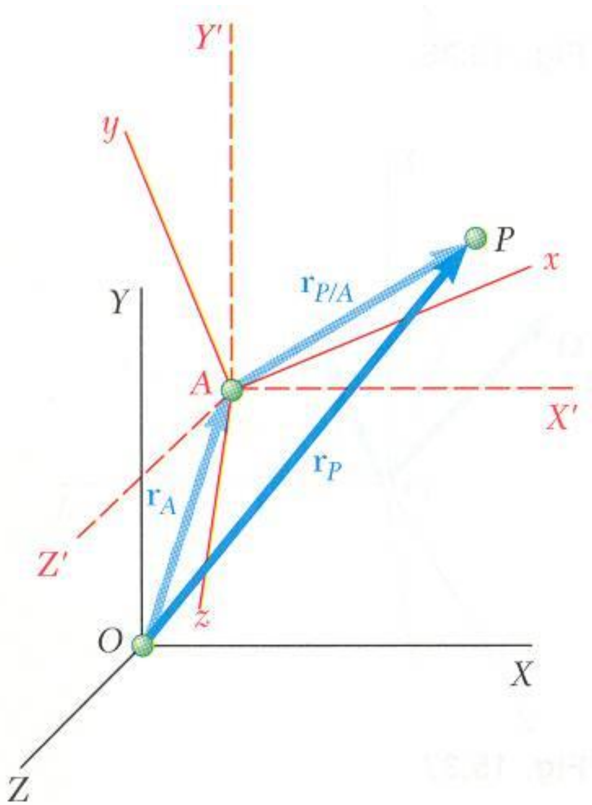
- The absolute acceleration can be expressed as

$$\vec{a}_P = \dot{\vec{\Omega}} \times \vec{r} + \vec{\Omega} \times \left(\vec{\Omega} \times \vec{r}\right) + 2\vec{\Omega} \times \left(\dot{\vec{r}}\right)_{Oxyz} + \left(\ddot{\vec{r}}\right)_{Oxyz}$$

$$= \vec{a}_{p'} + \vec{a}_{P/\mathscr{F}} + \vec{a}_c$$

$$\vec{a}_c = 2\vec{\Omega} \times \left(\dot{\vec{r}}\right)_{Oxyz} = 2\vec{\Omega} \times \vec{v}_{P/\mathscr{F}} = \text{Coriolis acceleration}$$

Frame of Reference in General Motion



Consider:

- fixed frame *OXYZ*,
- translating frame *AX'Y'Z'*, and
- translating and rotating frame *Axyz* or $\mathfrak{F}$.

- With respect to *OXYZ* and *AX'Y'Z'*,

$$\vec{r}_P = \vec{r}_A + \vec{r}_{P/A}$$

$$\vec{v}_P = \vec{v}_A + \vec{v}_{P/A}$$

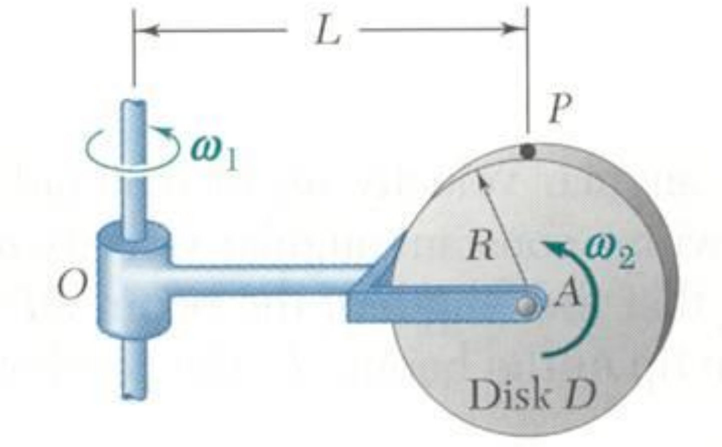
$$\vec{a}_P = \vec{a}_A + \vec{a}_{P/A}$$

- The velocity and acceleration of *P* relative to *AX'Y'Z'* can be found in terms of the velocity and acceleration of *P* relative to *Axyz*.

$$\vec{v}_P = \vec{v}_A + \vec{\Omega} \times \vec{r}_{P/A} + \left(\dot{\vec{r}}_{P/A}\right)_{Axyz}$$
$$= \vec{v}_{P'} + \vec{v}_{P/\mathfrak{F}}$$

$$\vec{a}_P = \vec{a}_A + \dot{\vec{\Omega}} \times \vec{r}_{P/A} + \vec{\Omega} \times \left(\vec{\Omega} \times \vec{r}_{P/A}\right)$$
$$+ 2\vec{\Omega} \times \left(\dot{\vec{r}}_{P/A}\right)_{Axyz} + \left(\ddot{\vec{r}}_{P/A}\right)_{Axyz}$$
$$= \vec{a}_{P'} + \vec{a}_{P/\mathfrak{F}} + \vec{a}_c$$

Sample Problem 10



For the disk mounted on the arm, the indicated angular rotation rates are constant.

Determine:

- the velocity of the point *P*,
- the acceleration of *P*, and
- angular velocity and angular acceleration of the disk.

<u>SOLUTION</u>:

- Define a fixed reference frame *OXYZ* at *O* and a moving reference frame *Axyz* or $\mathcal{F}$ attached to the arm at *A*.
- With *P'* of the moving reference frame coinciding with *P*, the velocity of the point *P* is found from

$$\vec{v}_P = \vec{v}_{P'} + \vec{v}_{P/\mathcal{F}}$$

- The acceleration of *P* is found from

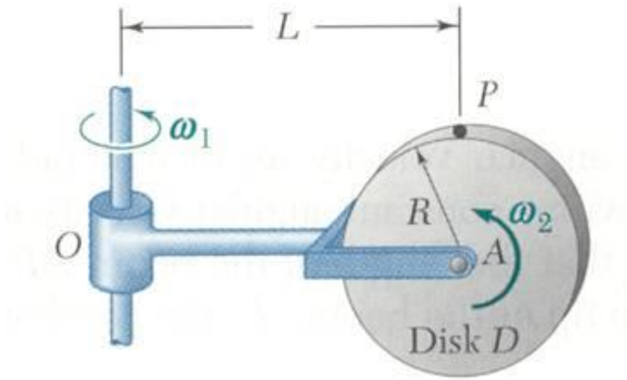
$$\vec{a}_P = \vec{a}_{P'} + \vec{a}_{P/\mathcal{F}} + \vec{a}_c$$

- The angular velocity and angular acceleration of the disk are

$$\vec{\omega} = \vec{\Omega} + \vec{\omega}_{D/\mathcal{F}}$$

$$\vec{\alpha} = \left(\dot{\vec{\omega}}\right)_{\mathcal{F}} + \vec{\Omega} \times \vec{\omega}$$

Sample Problem 10

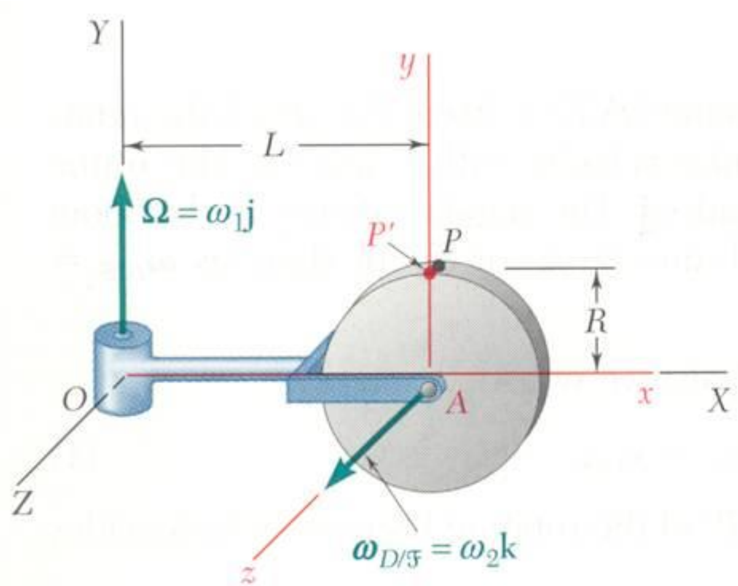


SOLUTION:

- Define a fixed reference frame *OXYZ* at *O* and a moving reference frame *Axyz* or $\mathfrak{F}$ attached to the arm at *A*.

$$\vec{r} = L\vec{i} + R\vec{j} \qquad \vec{r}_{P/A} = R\vec{j}$$

$$\vec{\Omega} = \omega_1 \vec{j} \qquad \vec{\omega}_{D/\mathfrak{F}} = \omega_2 \vec{k}$$



- With *P*′ of the moving reference frame coinciding with *P*, the velocity of the point *P* is found from

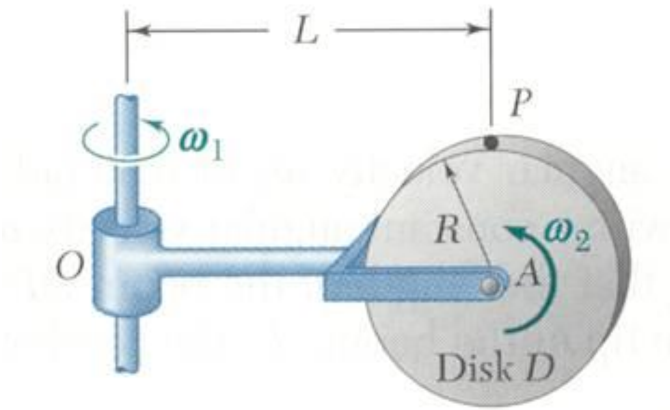
$$\vec{v}_P = \vec{v}_{P'} + \vec{v}_{P/\mathfrak{F}}$$

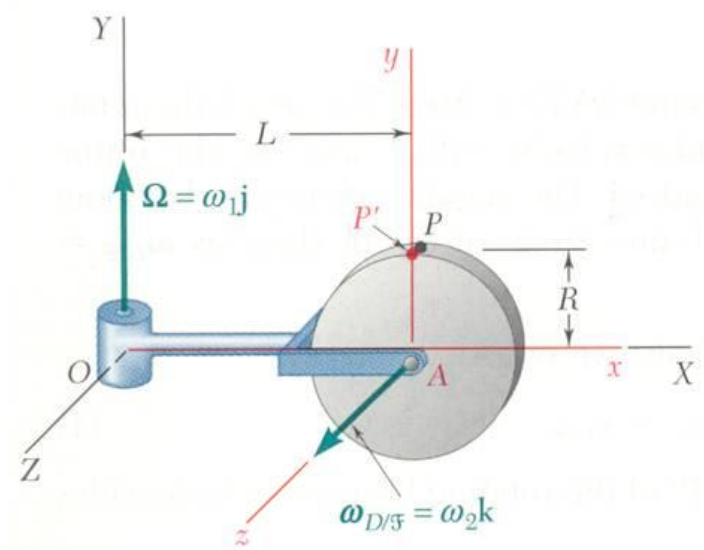
$$\vec{v}_{P'} = \vec{\Omega} \times \vec{r} = \omega_1 \vec{j} \times \left(L\vec{i} + R\vec{j}\right) = -\omega_1 L\,\vec{k}$$

$$\vec{v}_{P/\mathfrak{F}} = \vec{\omega}_{D/\mathfrak{F}} \times \vec{r}_{P/A} = \omega_2 \vec{k} \times R\vec{j} = -\omega_2 R\,\vec{i}$$

$$\boxed{\vec{v}_P = -\omega_2 R\,\vec{i} - \omega_1 L\,\vec{k}}$$

Sample Problem 10





- The acceleration of P is found from

$$\vec{a}_P = \vec{a}_{P'} + \vec{a}_{P/\mathcal{F}} + \vec{a}_c$$

$$\vec{a}_{P'} = \vec{\Omega} \times \left(\vec{\Omega} \times \vec{r} \right) = \omega_1 \vec{j} \times \left(-\omega_1 L \vec{k} \right) = -\omega_1^2 L \vec{i}$$

$$\vec{a}_{P/\mathcal{F}} = \vec{\omega}_{D/\mathcal{F}} \times \left(\vec{\omega}_{D/\mathcal{F}} \times \vec{r}_{P/A} \right)$$

$$= \omega_2 \vec{k} \times \left(-\omega_2 R \vec{i} \right) = -\omega_2^2 R \vec{j}$$

$$\vec{a}_c = 2\vec{\Omega} \times \vec{v}_{P/\mathcal{F}}$$

$$= 2\omega_1 \vec{j} \times \left(-\omega_2 R \vec{i} \right) = 2\omega_1 \omega_2 R \vec{k}$$

$$\boxed{\vec{a}_P = -\omega_1^2 L \vec{i} - \omega_2^2 R \vec{j} + 2\omega_1 \omega_2 R \vec{k}}$$

- Angular velocity and acceleration of the disk,

$$\vec{\omega} = \vec{\Omega} + \vec{\omega}_{D/\mathcal{F}} \qquad \boxed{\vec{\omega} = \omega_1 \vec{j} + \omega_2 \vec{k}}$$

$$\vec{\alpha} = \left(\dot{\vec{\omega}} \right)_{\mathcal{F}} + \vec{\Omega} \times \vec{\omega}$$

$$= \omega_1 \vec{j} \times \left(\omega_1 \vec{j} + \omega_2 \vec{k} \right) \qquad \boxed{\vec{\alpha} = \omega_1 \omega_2 \vec{i}}$$